

Chapter 7

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7.1 IC packages & Pin Diagrams of PIC18 Devices

Microchip Technology Inc, the manufacturer of PIC18 devices provides variety of packages based on the industrial and other needs. Some of the IC packages are presented here.

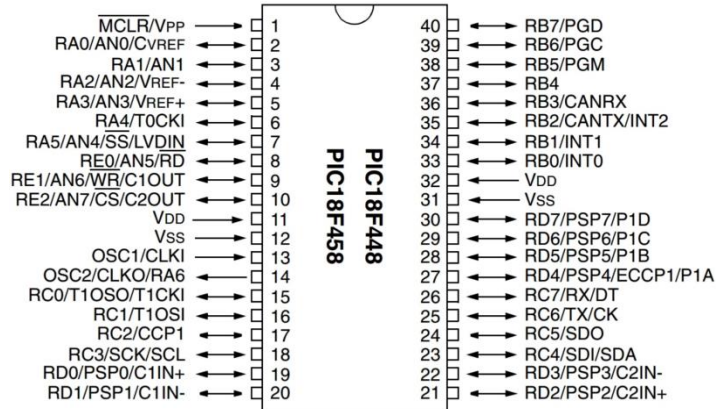


Fig 7.1a) PIC18F458 PDIP – Plastic Dual Inline Package with 40 Pins

Source: Microchip Document: PIC18FXX8 Data Sheet - DS41159E

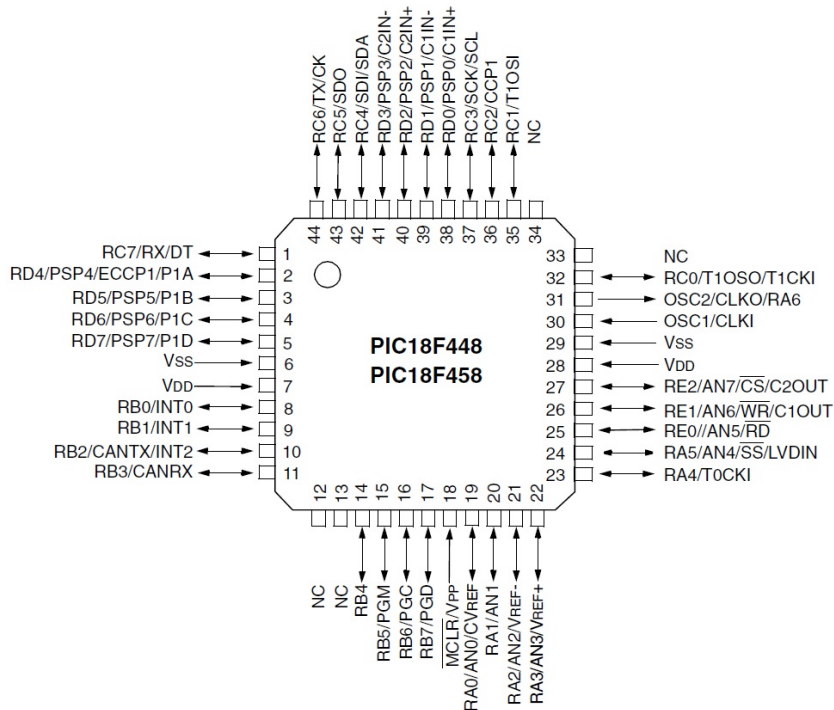


Fig 7.1b) PIC18F458 TQFP – Thin Quad Flat Package with 44 pins

Source: Microchip Document: PIC18FXX8 Data Sheet - DS41159E

The PIC18F458 is also available in multiple packages. They are

- PDIP – Plastic Dual Inline Package with 40 Pins, shown in fig 7.1a
- PLCC – Plastic Leaded Chip Carrier with 44 pins
- TQFP – Thin Quad Flat Package with 44 pins shown in fig 7.1b
- SPDIP, SOIC – Shrink Plastic Dual Inline Package, Small Outline Integrated Circuit with 40 pins

The PIC18C801 is available in two packages.

- TQFP – Thin Quad Flat Package with 80 pins, shown in fig 7.2
- PLCC – Plastic Leaded Chip Carrier with 84 pins

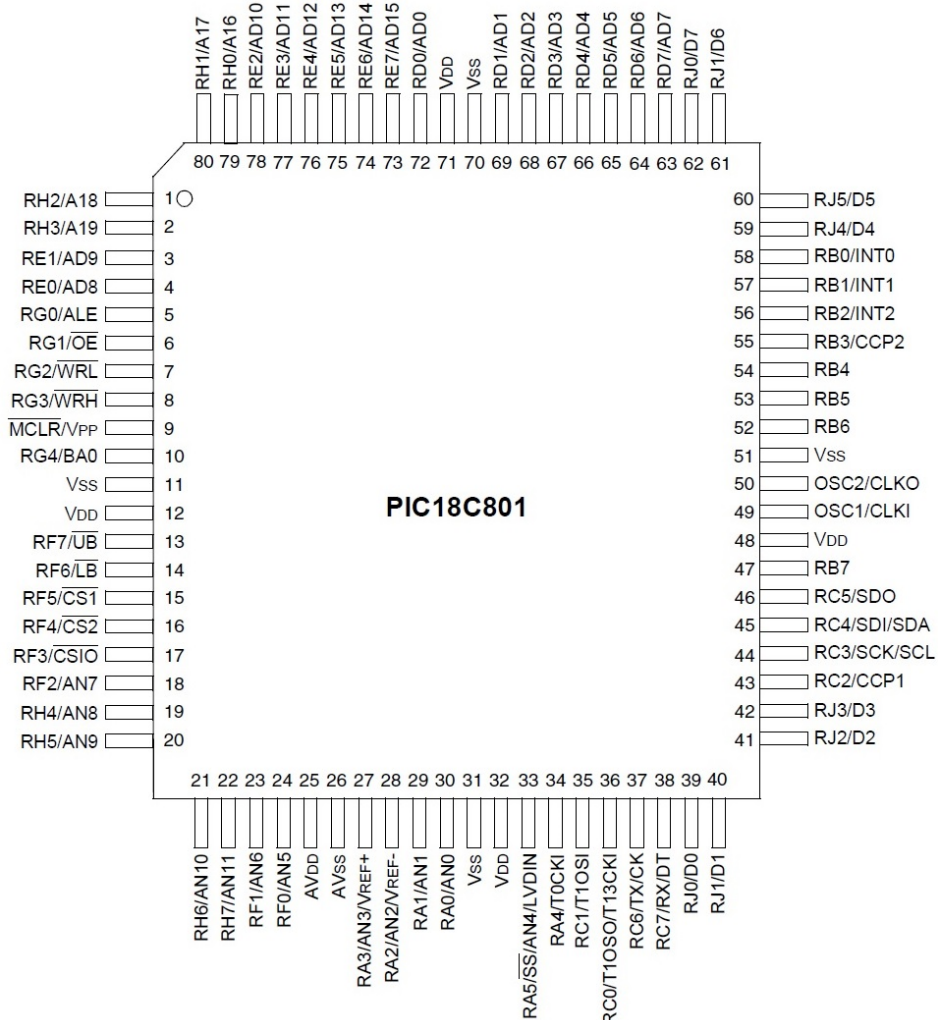


Fig 7.2 PIC18C801 TQFP – Thin Quad Flat Package with 80 pins

Source: Microchip Document: PIC18C601/801 Data Sheet - DS39541B

The PIC18F8720 is available in TQFP – Thin Quad Flat Package with 80 pins, shown in fig 7.3

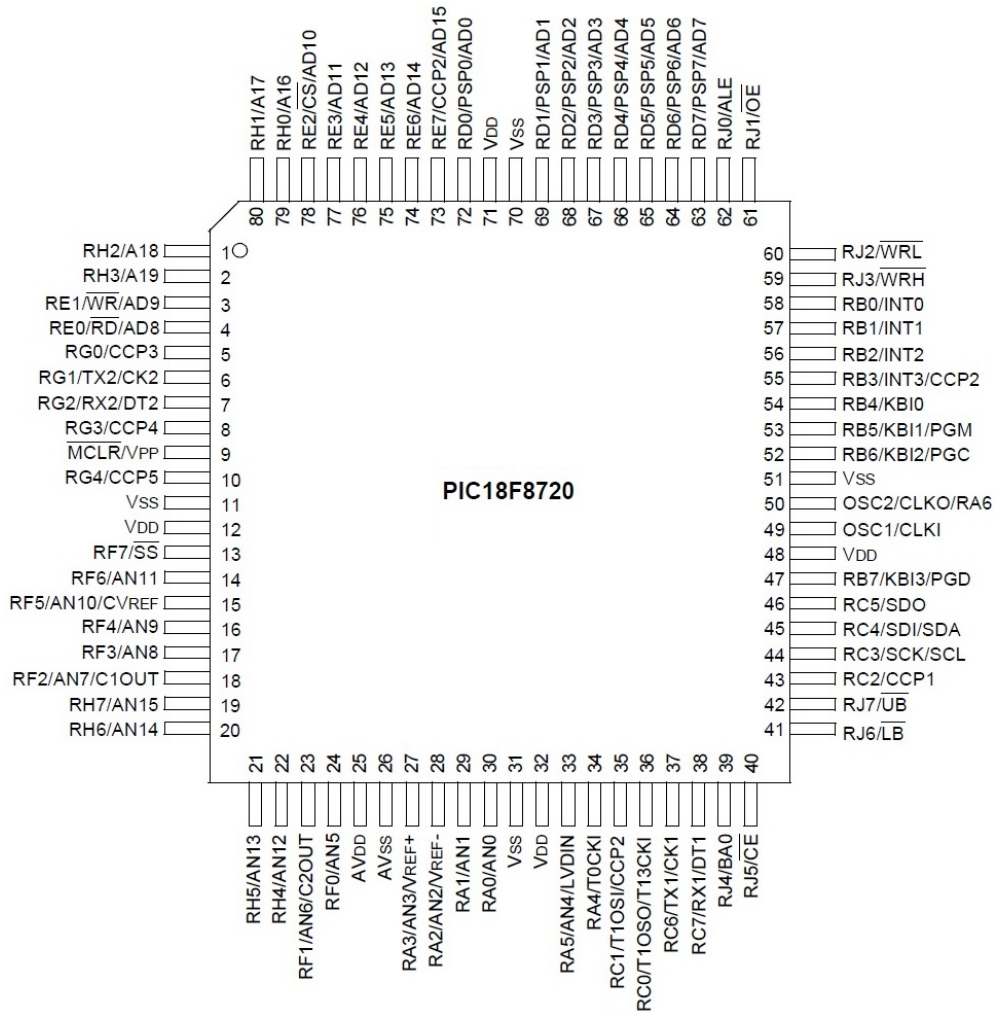


Fig 7.3 PIC18F8720 TQFP – Thin Quad Flat Package with 80 pins

Source: Microchip Document: PIC18F8720 Data Sheet - DS39609C

The pin functionalities are presented in detail in the respective peripheral sections.

7.2 I/O Port Structure

As discussed in the earlier section, the PIC18 devices vary based on their packages and number of pins. The number of available ports varies from one device to another and from one package to another. The PIC18F458 has 5 ports

Table 7.1: Summary of ports & I/O lines of PIC18F458, PIC18C801 & PIC18F8720

Port Details	PIC18F458	PIC18C801	PIC18F8720
Number of Ports	5	9	9
PORT Name	B, C, D, A(7), E(3)	B, C, D, E, F, H, J, A(6), G(5),	B, C, D, E, F, H, J, A(7), G(5),
Number of I/O Lines	34	67	68

For the effective use of given resources and keep the IC size to a minimum, the ports and pin outs in an IC are limited. Hence various signals from different peripherals are multiplexed together through a port pin to provide all the required signals external to the microcontroller.

The table 7.1 present the summary of different ports present in the PIC18 devices. Typically, all the ports if present (Port B, Port C, Port D, Port F, Port H & Port J) contain 8 pins each except for a few specific ports. In PIC18F458 the port A and port E are 7 and 3 pins each. The PIC18C801's port A and port G are 6 and 5 pins each and PIC18F8720 has 7 and 5 pins each for port A and port G. All the ports are bi-directional and are bit/byte addressable. It implies that all the port pins could be accessed (set/cleared or read) at once through a byte instruction or any specific pin/bit among the respective PORTx pins/bits can be accessed (set/cleared or read) separately using a bit-wise instruction.

A typical function of any port pin is to work as I/O - Input or Output. Additionally, a port pin may contain several other alternate functionalities related to different peripherals that are multiplexed into it. Various registers are provided for configuring the functionality available at a particular port pin. Some of these registers (TRISx, LATx) are for configuring the typical I/O functionality whereas some are for used for controlling the special functionalities of the peripherals (ADCONx, TxCON etc). The peripheral related registers are discussed in the respective sections. The three registers typically used for operating the port are given below.

- TRISx
- LATx
- PORTx

Here the 'x' represents the specific port and can take value 'A' to 'H' or 'J' depending on the actual port. The TRISx register is the Data Direction register that controls the direction of the PORTx. The TRISx register bits must be set or cleared to configure the PORTx as input or output. Clearing the specific TRISx bit to a 0 configures the respective PORTx pin as output.

Setting the TRISx bit to a 1 configures the respective PORTx pin as input. Upon reset or Power-up condition, all the port pins are set as inputs by default and are read as 0. This reduces power consumption as the port pins act as sink and not act as a source which happens when they are set as outputs.

When a peripheral is enabled using the peripheral related control registers, the typical I/O functionality is disabled, and the respective port pin provides the functionality corresponding to that peripheral. i.e. the special functions of PORTx override the general digital I/O function. As an example, ADC configuration overrides the digital I/O feature with an analog input feature and the I/O pin functionality of RB5 is overridden when low voltage ICSP serial programming mode is used.

Some peripherals override the pin functionality and change the pin that is configured as output to an input and vice versa, but it does not affect the value of TRIS bits. As an example, when Timer1 oscillator is enabled the TRISC<1:0> bits are ignored. Sometimes in addition to the special function feature, the port must be configured as input or output. As an example, PORTA pins used as analog inputs for ADC must also be configured as inputs using TRISA bits.

Before using any port in I/O mode, the respective port must be configured such that the direction of PORTx pins is set either as input or output. Once the direction is set, the PORTx may be used for the required I/O operations.

A read-modify-write operation is always performed when a file register is specified as part of any instruction. The respective register is read; the data is then modified and stored based on the chosen destination through the option 'd' specified as part of the instruction. A read operation is always performed even when the file register is being written because of the given instruction. The "CLR PORTB" instruction reads the PORTB, clears all the bits and then writes the result to PORTB.

When a port is read using the PORTx register, the status of the port pins is read. When a value is written to port using the PORTx register the value is written to ports data latch LATx register. The value that the I/O pins are driven is set by performing the read-modify write operation on data latch LATx register. When a Read-modify-write operation is performed on the LATx register, the data latch LATx is read and written to LATx register for the PORTx. In certain cases, where the last read value is desired or when a port read causes some event, the data latch LATx register must be read instead of PORTx.

Reading PORTx replaces the data latch LATx register with the new pin value. An example for change in event occurs when reading PORTB. PORTB has interrupt on change feature and reading PORTB clears a mismatch and clears the condition that sets the RBIF flag. Hence in such a case the subsequent reads must be reading of a latch rather than the port.

Table 7.2: Overview of PORT registers & its bits of PIC18F458, PIC18C801 & PIC18F8720

PORTA										
PIC18F458, PIC18F8720	Register Description	Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
	Direction Register	TRISA	—	TRISA6	TRISA5	TRISA4	TRISA3	TRISA2	TRISA1	TRISA0
	Data O/P Latch Register	LATA	—	LATA6	LATA5	LATA4	LATA3	LATA2	LATA1	LATA0
	PORT Register	PORTA	—	RA6	RA5	RA4	RA3	RA2	RA1	RA0
PIC18C801	Register Name	Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
	Direction Register	TRISA	—	—	TRISA5	TRISA4	TRISA3	TRISA2	TRISA1	TRISA0
	Data O/P Latch Register	LATA	—	—	LATA5	LATA4	LATA3	LATA2	LATA1	LATA0
	PORT Register	PORTA	—	—	RA5	RA4	RA3	RA2	RA1	RA0
PORTB										
PIC18F458, PIC18C801 & PIC18F8720	Register Description	Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
	Direction Register	TRISB	TRISB7	TRISB6	TRISB5	TRISB4	TRISB3	TRISB2	TRISB1	TRISB0
	Data O/P Latch Register	LATB	LATB7	LATB6	LATB5	LATB4	LATB3	LATB2	LATB1	LATB0
	PORT Register	PORTB	RB7	RB6	RB5	RB4	RB3	RB2	RB1	RB0
PORTC										
PIC18F458, PIC18C801 & PIC18F8720	Register Description	Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
	Direction Register	TRISC	TRISC7	TRISC6	TRISC5	TRISC4	TRISC3	TRISC2	TRISC1	TRISC0
	Data O/P Latch Register	LATC	LATC7	LATC6	LATC5	LATC4	LATC3	LATC2	LATC1	LATC0
	PORT Register	PORTC	RC7	RC6	RC5	RC4	RC3	RC2	RC1	RC0
PORTD										
PIC18F458, PIC18C801 & PIC18F8720	Register Description	Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
	Direction Register	TRISD	TRISD7	TRISD6	TRISD5	TRISD4	TRISD3	TRISD2	TRISD1	TRISD0
	Data O/P Latch Register	LATD	LATD7	LATD6	LATD5	LATD4	LATD3	LATD2	LATD1	LATD0
	PORT Register	PORTD	RD7	RD6	RD5	RD4	RD3	RD2	RD1	RD0
PORTE										
PIC18F458	Register Description	Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
	Direction Register	TRISE	IBF	OBF	IBOV	PSPMODE	—	TRISE2	TRISE1	TRISE0
	Data O/P Latch Register	LATE	—	—	—	—	—	LATE2	LATE1	LATE0
	PORT Register	PORTE	—	—	—	—	—	RE2	RE1	RE0
PIC18C801, PIC18F8720	Register Description	Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
	Direction Register	TRISE	TRISE7	TRISE6	TRISE5	TRISE4	TRISE3	TRISE2	TRISE1	TRISE0
	Data O/P Latch Register	LATE	LATE7	LATE6	LATE5	LATE4	LATE3	LATE2	LATE1	LATE0
	PORT Register	PORTE	RE7	RE6	RE5	RE4	RE3	RE2	RE1	RE0
PORTF										
PIC18C801, PIC18F8720	Register Description	Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
	Direction Register	TRISF	TRISF7	TRISF6	TRISF5	TRISF4	TRISF3	TRISF2	TRISF1	TRISF0
	Data O/P Latch Register	LATF	LATF7	LATF6	LATF5	LATF4	LATF3	LATF2	LATF1	LATF0
	PORT Register	PORTF	RF7	RF6	RF5	RF4	RF3	RF2	RF1	RF0
PORTG										
PIC18C801, PIC18F8720	Register Description	Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
	Direction Register	TRISG	—	—	—	TRISG4	TRISG3	TRISG2	TRISG1	TRISG0
	Data O/P Latch Register	LATG	—	—	—	LATG4	LATG3	LATG2	LATG1	LATG0
	PORT Register	PORTG	—	—	—	RG4	RG3	RG2	RG1	RG0
PORTH										
PIC18C801, PIC18F8720	Register Description	Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
	Direction Register	TRISH	TRISH7	TRISH6	TRISH5	TRISH4	TRISH3	TRISH2	TRISH1	TRISH0
	Data O/P Latch Register	LATH	LATH7	LATH6	LATH5	LATH4	LATH3	LATH2	LATH1	LATH0
	PORT Register	PORTH	RH7	RH6	RH5	RH4	RH3	RH2	RH1	RH0
PORTJ										
PIC18C801, PIC18F8720	Register Description	Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
	Direction Register	TRISJ	TRISJ7	TRISJ6	TRISJ5	TRISJ4	TRISJ3	TRISJ2	TRISJ1	TRISJ0
	Data O/P Latch Register	LATJ	LATJ7	LATJ6	LATJ5	LATJ4	LATJ3	LATJ2	LATJ1	LATJ0
	PORT Register	PORTJ	RJ7	RJ6	RJ5	RJ4	RJ3	RJ2	RJ1	RJ0

Table 7.3: Overview of PORT Pins of PIC18F458, PIC18C801 & PIC18F8720.

Port	Common Pin Function	PIC18F458	PIC18C801	PIC18F8720
		Additional Pin Function		
		Pin Name	Pin Name	Pin Name
PORT A	RA0 / AN0	CVREF		
	RA1 / AN1			
	RA2 / AN2 / VREF-			
	RA3 / AN3 / VREF+			
	RA4 / T0CKI			
	RA5 / AN4 / LVDIN	$\overline{SS}$	$\overline{SS}$	
		RA6 / OSC2 / CLKO		RA6 / OSC2 / CLKO
PORT B	RB0 / INT0			
	RB1 / INT1			
	RB2 / INT2	CANTX		
	RB3	CANRX	CCP2	INT3 / CCP2 (1)
	RB4			KBIO
	RB5	PGM		KB1 / PGM
	RB6	PGC		KB2 / PGC
	RB7	PGD		KB3 / PGD
PORT C	RC0 / T1OSO	T1CKI	T13CKI	T13CKI
	RC1 / T1OSI			CCP2 (1)
	RC2 / CCP1			
	RC3 / SCK / SCL			
	RC4 / SDI / SDA			
	RC5 / SDO			
	RC6	TX / CK	TX / CK	TX1 / CK1
	RC7	RX / DT	RX / DT	RX1 / DT1
PORT D	RD0	PSP0 / C1IN+	AD0 / A0 (2)	PSP0 / AD0
	RD1	PSP1 / C1IN-	AD1 / A1 (2)	PSP1 / AD1
	RD2	PSP2 / C2IN+	AD2 / A2 (2)	PSP2 / AD2
	RD3	PSP3 / C2IN-	AD3 / A3 (2)	PSP3 / AD3
	RD4	PSP4 / P1A / ECCP1	AD4 / A4 (2)	PSP4 / AD4
	RD5	PSP5 / P1B	AD5 / A5 (2)	PSP5 / AD5
	RD6	PSP6 / P1C	AD6 / A6 (2)	PSP6 / AD6
	RD7	PSP7 / P1D	AD7 / A7 (2)	PSP7 / AD7
PORT E	RE0	AN5 / $\overline{RD}$	AD8 / A8 (2)	$\overline{RD}$ / AD8
	RE1	AN6 / $\overline{WR}$ / C1OUT	AD9 / A9 (2)	$\overline{WR}$ / AD9
	RE2	AN7 / $\overline{CS}$ / C2OUT	AD10 / A10 (2)	$\overline{CS}$ / AD10
			RE3 / AD11 / A11 (2)	RE3 / AD11
			RE4 / AD12 / A12 (2)	RE4 / AD12
			RE5 / AD13 / A13 (2)	RE5 / AD13
			RE6 / AD14 / A14 (2)	RE6 / AD14
			RE7 / AD15 / A15 (2)	RE7 / AD15 / CCP2 (1)
PORT F			RF0 / AN5	RF0 / AN5
			RF1 / AN6	RF1 / AN6 / C2OUT
			RF2 / AN7	RF2 / AN7 / C1OUT
			RF3 / $\overline{CSIO}$	RF3 / AN8 / C2IN+
			RF4 / A16 / $\overline{CS2}$	RF4 / AN9 / C2IN-
			RF5 / $\overline{CS1}$	RF5 / AN10 / CVREF / C1IN+
			RF6 / $\overline{LB}$	RF6 / AN11 / C1IN-
			RF7 / $\overline{UB}$	RF7 / $\overline{SS}$

Port	Common Pin Function	PIC18F458	PIC18C801	PIC18F8720
		Additional Pin Function		
		Pin Name	Pin Name	Pin Name
PORT G			RG0 / ALE	RG0 / CCP3
			RG1 / $\overline{OE}$	RG1 / TX2 / CK2
			RG2 / WRL	RG2 / RX2 / DT2
			RG3 / WRH	RG3 / CCP4
			RG4 / BA0	RG4 / CCP5
PORT H			RH0 / A16	RH0 / A16
			RH1 / A17	RH1 / A17
			RH2 / A18	RH2 / A18
			RH3 / A19	RH3 / A19
			RH4 / AN8	RH4 / AN12
			RH5 / AN9	RH5 / AN13
			RH6 / AN10	RH6 / AN14
PORT J			RH7 / AN11	RH7 / AN15
			RJ0 / D0	RJ0 / ALE
			RJ1 / D1	RJ1 / $\overline{OE}$
			RJ2 / D2	RJ2 / $\overline{WRL}$
			RJ3 / D3	RJ3 / $\overline{WRH}$
			RJ4 / D4	RJ4 / BA0
			RJ5 / D5	RJ5 / $\overline{CE}$
			RJ6 / D6	RJ6 / $\overline{LB}$
			RJ7 / D7	RJ7 / $\overline{UB}$

Refer to Notes of Table 7.4 for notes 1 and 2 of Table 7.3.

The table 7.2 presents the registers and associated bits for various ports. The table 7.3 presents the overview of PORT pins for the PIC18F458, PIC18C801 & PIC18F8720 devices with Column 2 containing the common pin functionality across all the three devices. The columns 3, 4 & 5 contain the respective devices additional functionalities.

7.2.1 Overview of Functional Modules & its Pins

The table 7.3 approach allows for easy overview of pin functions with respect to a specific port. The table 7.4 presents the categorization of all the pins, overview on availability of module/functions and presents the functional module wise overview with their port pins for the three devices. A brief overview of the same is presented here and details in respective sections.

To begin with, the PIC18F458 has 5 ports PORTA (7), PORTB (8), PORTC (8), PORTD (8) and PORTE (3) with RA7 and RE3:R7 missing pins. The PIC18C801 has 9 ports PORTA (6), PORTB (8), PORTC (8), PORTD (8), PORTE (8), PORTF (8), PORTG (5), PORTH (8) and PORTJ (8) with RA6, RA7 and RG5:RG7 missing pins. The PIC18F8720 has 9 ports PORTA (7), PORTB (8), PORTC (8), PORTD (8), PORTE (8), PORTF (8), PORTG (5), PORTH (8) and PORTJ (8) with RA7 and RG5:RG7 missing pins. The number in the parenthesis indicating the number of pins present in that port.

Table 7.4: Module/Function wise overview of PORT Pins of PIC18 F458, C801 & F8720

PORT/Pin Name	Pin Notation	PIC18F458	PIC18C801	PIC18F8720	Note
		PORT Pin	PORT Pin	PORT Pin	
PORTS					
PORT A	RA0:RA5	Y	Y	Y	-
	RA6	Y	-	Y	4
PORT B	RB0:RB7	Y	Y	Y	-
PORT C	RC0:RC7	Y	Y	Y	-
PORT D	RD0:RD7	Y	Y	Y	-
PORT E	RE0:RE2	Y	Y	Y	-
	RE3:RE7	-	Y	Y	3
PORT F	RF0:RF7	-	Y	Y	3
PORT G	RG0:RG4	-	Y	Y	3
PORT H	RH0:RH7	-	Y	Y	3
PORT J	RJ0:RJ7	-	Y	Y	3
External Interrupts					
PORTB Change Input	RB4:RB7	RB4:RB7	RB4:RB7	KBIO:KBIO3	3, 4, 6
External Interrupt Inputs	INT0:INT2	RB0:RB2	RB0:RB2	RB0:RB2	-
	INT3	-	-	RB3	3, 4
Oscillators					
Oscillator Input	OSC1	Y	Y	Y	-
Oscillator Output	OSC2	RA6	Y	RA6	-
Clock Output	CLKO	RA6	Y	RA6	-
Clock Input	CLKI	Y	Y	Y	-
Timers					
Timer 0 Clock Input	T0CKI	RA4	RA4	RA4	
Timer 1 Clock Input	T1CKI	RC0, T1CKI	RC0, T13CKI	RC0, T13CKI	6
T1 Oscillator Output	T1OSO	RC0	RC0	RC0	-
T1 Oscillator Input	T1OSI	RC1	RC1	RC1	-
E-Enhanced, CCP-Capture/Compare/PWM-Pulse Width Modulation					
CCP1 Input/Output	CCP1 (8)	RC2	RC2	RC2	-
CCP2 Input/Output	CCP2 (8)	-	RB3	RB3/RC1/RE7	1, 3, 7
CCP3 Input/Output	CCP3 (8)	-	-	RG0	3, 4
CCP4 Input/Output	CCP4 (8)	-	-	RG3	3, 4
CCP5 Input/Output	CCP5 (8)	-	-	RG4	3, 4
ECCP1 Input/Output	ECCP1 (8)	RD4	-	-	4, 5
ECCP1 PWM Out1	P1A	RD4	-	-	4, 5
ECCP1 PWM Out2	P1B	RD5	-	-	4, 5
ECCP1 PWM Out3	P1C	RD6	-	-	4, 5
ECCP1 PWM Out3	P1D	RD7	-	-	4, 5
ADC: Analog to Digital Converter					
Analog Input 0:3	AN0:AN3	RA0:RA3	RA0:RA3	RA0:RA3	-
Analog Input 4	AN4	RA5	RA5	RA5	-
Analog Input 5:7	AN5:7	RE0:RE2	RF0:RF2	RF0:RF2	7
Analog Input 8:11	-	RH4:RH7	RH4:RH7	RF3:RF6	3, 7
Analog Input 12:15	-	-	-	RH4:RH7	3, 4
Voltage Reference					
Voltage Input Reference-	VREF-	RA2	RA2	RA2	-
Voltage Input Reference+	VREF+	RA3	RA3	RA3	-

PORT/Pin Name	Pin Notation	PIC18F458	PIC18C801	PIC18F8720	Note
		PORT Pin	PORT Pin	PORT Pin	
Comparator Voltage Reference					
Comparator Reference Voltage Output	CVREF	RA0	-	RF5	4, 7
Comparator					
Comparator 1 Non Inv Input	C1IN+	RD0	-	RF5	4, 7
Comparator 1 Inv Input	C1IN-	RD1	-	RF6	4, 7
Comparator 1 Output	C1OUT	RE1	-	RF2	4, 7
Comparator 2 Non Inv Input	C2IN+	RD2	-	RF3	4, 7
Comparator 2 Inv Input	C2IN-	RD3	-	RF4	4, 7
Comparator Output	C2OUT	RE2	-	RF1	4, 7
PSP - Parallel Slave Port					
PSP Data Lines Input/Output	PSP0:PSP7	RD0:RD7	-	RD0:RD7	4
Read Output	$\overline{RD}$	RE0	-	RE0	4
Write Output	$\overline{WR}$	RE1	-	RE1	4
Chip Select	$\overline{CS}$	RE2	-	RE2	4
Addressable USART					
Serial Transmit 1	TX1	RC6, TX	RC6, TX	RC6, TX1	6
Serial Receive 1	RX1	RC7, RX	RC7, RX	RC7, RX1	6
Serial Transmit 2	TX2	-	-	RG1, TX2	3, 4
Serial Receive 2	RX2	-	-	RG2, RX2	3, 4
Synchronous Clock1	CK1	RC6, CK	RC6, CK	RC6, CK1	6
Serial Data1	DT1	RC7, DT	RC7, DT	RC7, DT1	6
Synchronous Clock2	CK2	-	-	RG1, CK2	3, 4
Serial Data2	DT2	-	-	RG2, DT2	3, 4
MSSP - Master Synchronous Serial Port : I2C - Inter Integrated Circuit					
Serial Clock	SCL	RC3	RC3	RC3	-
Serial Data	SDA	RC4	RC4	RC4	-
MSSP - Master Synchronous Serial Port : SPI - Serial Peripheral Interface					
Serial Clock	SCK	RC3	RC3	RC3	-
Serial Data In	SDI	RC4	RC4	RC4	-
Serial Data Out	SDO	RC5	RC5	RC5	-
Slave Select	$\overline{SS}$	RA5	RA5	RF7	7
CAN - Controller Area Network					
CAN Transmit	CANTX	RB2	-	-	4, 5
CAN Receive	CANRX	RB3	-	-	4, 5
Low Voltage Detect					
Low Voltage Detect Input	LVDIN	RA5	RA5	RA5	-
ICSP - In-Circuit Serial Programming					
Low-voltage ICSP programming Enable	PGM	RB5	-	RB5	4
ICSP programming Clock	PGC	RB6	RB6 *	RB6	4
ICSP programming Data	PGD	RB7	RB7 *	RB7	4

PORT/Pin Name	Pin Notation	PIC18F458	PIC18C801	PIC18F8720	Note
		PORT Pin	PORT Pin	PORT Pin	
EMI - External Memory Interface					
Data Lines, 8-Bit Mode (2)	D0:D7	-	RJ0:RJ7	-	2, 3, 5
Address Lines, 8-Bit Mode (2)	A0:A7	-	RD0:RD7	-	2, 3, 5
	A8:A15	-	RE0:RE7	-	2, 3, 5
Address Lines, 8/16-Bit Mode	A16	-	RF4 / RH0	RH0	2, 3, 7
	A17:A19	-	RH1:RH3	RH1:RH3	2, 3
Address/Data Lines, 16-Bit Mode (2)	AD0:AD7	-	RD0:RD7	RD0:RD7	2, 3
	AD8:AD15	-	RE0:RE7	RE0:RE7	2, 3
System Bus Chip Select I/O	CSIO	-	RF3	-	3, 5, 7
Chip Select 1	CS1	-	RF5	-	3, 5, 7
Chip Select 2	CS2	-	RF4	-	3, 5, 7
Low Byte Select Control Signal for EMI	$\overline{LB}$	-	RF6	RJ6	3, 7
High Byte Select Control Signal for EMI	$\overline{UB}$	-	RF7	RJ7	3, 7
Address Latch Enable Control for EMI	ALE	-	RG0	RJ0	3, 7
Output Enable Control for EMI	$\overline{OE}$	-	RG1	RJ1	3, 7
Write Low Byte Control for EMI	$\overline{WRL}$	-	RG2	RJ2	3, 7
Writ High Byte Control for EMI	$\overline{WRH}$	-	RG3	RJ3	3, 7
System Bus Byte Address 0 Control for EMI	BA0	-	RG4	RJ4	3, 7
Chip Enable Control for EMI	$\overline{CE}$	-	-	RJ5	3, 4

Note	
1	RB3: Alternate assignment for CCP2 when CCP2MX is not selected (all operating modes) RC1: Default assignment when CCP2MX is set. RE7: Alternate assignment for CCP2 when CCP2MX is not selected (all operating modes) CCP2MX: CCP2 Mux bit In Microcontroller mode: 1 = CCP2 input/output is multiplexed with RC1 0 = CCP2 input/output is multiplexed with RE7 In Microprocessor, Microprocessor with Boot Block and Extended Microcontroller 1 = CCP2 input/output is multiplexed with RC1 0 = CCP2 input/output is multiplexed with RB3
2	For PIC18C801 In 16-Bit mode the pins are ADx multiplexed address/data bus, In 8-Bit
3	Not in PIC18F458
4	Not in PIC18C801
5	Not in PIC18F8720
6	Bit name varies across devices
7	Bit position varies by port pin across devices
8	In Capture Mode the pin is input, In compare or PWM mode the pin is output

External Interrupts: RB4:RB7, INT0:INT3

There are two types of external interrupts. 1) Port B change or Interrupt on change on all the three devices, 2) External interrupts INT0 to INT2 on all the three devices and external interrupt INT3 additionally on PIC18F8720.

The PORTB change or Interrupt on change interrupt is available on pins RB4:RB7. In PIC18F8720 the PORTB change interrupt pins are named as KBI0:KBI3. An event is considered to have occurred when a change of value occurs at any one of pins RB4:RB7 and an interrupt is generated for any such event. The INT0:INT3 are available on pins RB0:RB3. These are specifically called external interrupts and may be configured to be triggered based on positive or negative edges. The details of these are presented in external interrupts section 7.6.

Oscillator/Clock: OSC1, OSC2, CLKI & CLKO

The PIC18 devices could be operated in one of eight (four) oscillator modes by configuring the FOSC2, FOSC1 & FOSC0 bits of CONFIG1H register. The CONFIG1H register of PIC18C801 does not contain FOSC2 bit and hence is operable in only first 4 modes. The PIC18F458 & PIC18F8720 are operable in all 8 oscillator modes.

- 1) LP: Low Power Crystal
- 2) HS: High Speed Crystal/Resonator
- 3) RC: External Resistor/Capacitor and
- 4) EC: External Clock.
- 5) XT: Crystal/Resonator
- 6) HS4: High-Speed Crystal/Resonator with PLL enabled
- 7) RCIO: External Resistor/Capacitor with I/O pin enabled
- 8) ECIO: External Clock with I/O pin enabled

When the PIC18 is connected with an oscillator, i.e. in modes 1, 2, 5 and 6 the two oscillator pins OSC1 & OSC2 are used. The OSC2 is multiplexed with I/O pin functionality of RA6 in F458 & F8720 whereas in C801 RA6 is not available and is dedicated for OSC2. When the PIC18 is connected in the other modes i.e. in modes 3, 4, 7 and 8 the clock input is provided on CLKI pin and clock out ($f_{osc}/4$) may be obtained from CLKO. When the clock output signal is not used, the pin RA6 could be used as an I/O pin in F458 & F8720. The modes and details of clock configurations are presented in the special features section of chapter 16.

Timers: T0CKI, T1CKI (T13CKI), T1OSI & T1OSO

The PIC18 devices are provided with 4 or 5 internal timers. Of the 4/5 timers, the Timer0, Timer 1 and Timer 3 could be made to operate based on external clock input signals T0CKI for Timer 0, and T1CKI (T13CKI) for Timer1/Timer3. Timer 1 and Timer 3 may be

operated by a crystal oscillator connected between T1OSI and T1OSO pins. Timer 0 is not provided the flexibility of being connected with an oscillator and hence can only be connected by a clock input T0CKI.

The Timer 3 is not provided with separate clock input or oscillator pins. The oscillator connection pins T1OSI & T1OSO and clock input pin T1CKI (T13CKI) is common for both the Timer 1 and Timer 3. In PIC18F458 it is just named as T1CKI and in PIC18C801 and PIC18F8720 it is named as T13CKI to represent commonality between Timer1 and Timer3. When the timers work as counters, the inputs for the counter are provided at the T0CKI or T1CKI (T13CKI) pins. The details of the timers are presented in timer's section in chapter 9.

CCP/ECCP – Capture Compare/Enhanced Capture Compare: CCP1:CCP5, P1A:P1D.

The number of CCP – Capture Compare modules vary from 1 to 5 based on the PIC18 device. The PIC18F458 contains only CCP1, PIC18C801 has CCP1 and CCP2 where as PIC18F8720 contains 5 CCP's i.e. CCP1, CCP2, CCP3, CCP4 and CCP5. Each module is provided with a pin function named as CCP1, CCP2, CCP3, CCP4 and CCP5 respectively. The CCPx pin operates as an input or an output.

When the CCP module is configured for capture, the pin must be operated as an input so that signal that must be captured is applied at the input. When the module is captured as compare or PWM – Pulse Width Modulation the pin is may be configured as output to obtain the status of compare or PWM signals.

The ECCP – Enhanced Capture Compare module is only present in PIC18F458. It provides for enhanced PWM features like the full bridge or half bridge outputs that are not present in regular CCP modules. Hence when the ECCP module is configured in PWM mode the P1A, P1B, P1C and P1D serve as the enhanced PWM output pins to provide full bridge or half bridge output signals. The details of the CCP/ECCP modules are presented in CCP/ECCP/PWM section in chapter 13.

ADC: Analog to Digital Converter: AN0:AN15

All the three PIC18 devices are 10-Bit ADCs with PIC18F458 having 8, PIC18C801 having 12 and PIC18F8720 having 16 analog inputs respectively. The ADC converts an analog signal value into a digital value. The analog input for the ADC is provided through the analog input pins named as AN0:AN15. These pins must be configured as inputs.

The three PIC18 devices presented mainly vary based on the number of input channels and the names associated with the register bits and their locations along with analog input port pins. The variation of the respective port pins for the corresponding analog inputs is presented in table 7.4. The details of the ADC modules are presented in ADC section in chapter 12.

Voltage Reference: VREF- & VREF+

All the three PIC18 devices are equipped to take +ve and –ve reference voltage V_{REF+} & V_{REF-} as external input signal for modules such as ADC, comparator etc. So, the corresponding voltages may be applied at these pins for use by the modules such as ADC, comparator etc. These are provided through the RA2 and RA3 port pins and configured through the ADC control registers. The details of the configuration for applying the voltage reference signals as input is presented in ADC section in chapter 12.

Comparators & Comparator Voltage Reference: C1IN+, C2IN+, C1IN-, C2IN-, C1OUT, C2OUT & CVREF

The PIC18F458 and PIC18F8720 are provided with 2 comparators: Comparator 1 and Comparator 2 but has only one common flag CMIF and interrupt. Each comparator 1 & 2 has a Noninverting and inverting input that is applied to an op-amp based comparator. The Noninverting and inverting inputs are applied through the two external input pins CxIN+ and CxIN-. The output of each comparator is also routed to the external physical pin CxOUT. Here x represents 1 or 2 indicating comparator 1 and comparator 2.

The PIC18F458 and PIC18F8720 are also provided with one internal voltage reference module whose voltage may be provided as reference voltage to external devices through the CVREF function pin. The comparators may also be applied with the input reference provided through V_{REF+} & V_{REF-} pins and are configured through the ADC control registers. The details of the configuration for applying the voltage reference signals as input is presented in ADC section in chapter 12.

PIC18C801 does not contain comparator or the voltage reference module. The two PIC18 devices with comparators mainly vary based on the port pins used and are otherwise identical. The details of the comparators & its configuration are presented in Comparators section in chapter 12.

PSP: Parallel Slave Port: PSP0:PSP7, RD, WR, CS

Only the PIC18F458 and PIC18F8720 have the feature of PSP – Parallel Slave Port. The parallel slave port feature is used for parallel data transfer and is similar to that of traditional Programmable Peripheral Interface 8255 but is not the same.

PSP is mainly used for parallel communication with control signals. Data is exchanged on PORTD whereas control signals are available through the PORTE. The Table 7.4 presents the actual pins related to the PSP. The Read, Write of PSP & Chip select signal to enable the parallel device is available through the pin function RD, WR & CS of PORTE. The details of the PSP module are presented in Interfacing of PSP section in chapter 10.

Addressable USART: Universal Synchronous Asynchronous Receive Transmit: TX (TX1), RX (RX1), TX2, RX2, CK (CK1), DT (DT1), CK2 & DT2

The USART- Universal Synchronous Asynchronous Receive Transmit is also known as SCI: Serial Communications interface. It is another module that is used for serial communication and is available in all the three devices. The PIC18F8720 has two independently configurable USARTs while the other two PIC18 devices have only one. The basic functionality is identical in all the three devices except for the port pins and the function name that is used.

When USART is configured for Asynchronous communication, it is operated in full duplex mode, and the serial data is transmitted on TX pin and data is received on RX pin using the asynchronous serial communication protocol. Since PIC18F8720 has two USARTs, two serial transmit function pins TX1 and TX2, two serial receive function pins RX1 and RX2 are present in it. The PIC18F458 and PIC18C801 have only one USART and hence have the TX and RX function pins.

The addressable USART feature supports address detection and enables interrupt on the detection of address bit.

When USART is configured for Synchronous communication, it is operated in half duplex mode, and the serial data is transmitted or received on DT pin and the clock is transmitted or received on CK pin. Since PIC18F8720 has two USARTs, two data transfer function pins DT1 and DT2, two clock function pins RX1 and RX2 are present in it. The PIC18F458 and PIC18C801 have only one USART and hence have the DT and CK function pins. The details of the USART module are presented in USART section of chapter 10.

MSSP: Master Synchronous Serial Port - I2C: Inter Integrated Circuit Protocol: SCL & SDA

The MSSP: Master Synchronous Serial Port is a serial communication module that could be configured to communicate using I2C or SPI serial protocols. All the three PIC18 devices support I2C serial communication protocol. The PIC18 device could also be configured as a master or slave and can either transmit or receive the data and can generate clock or operate based on clock provided by external master. When MSSP is configured for I2C the serial clock and serial data are transmitted or received on SCL and SDA function pins. Since the I2C operates in Half Duplex mode, data transmitted or received takes place on SDA line. The transmission or reception of data or clock is through the same SDA or SCL function pins. PIC18 devices support 7-bit or 10-bit addressing mechanism. The details of the MSSP module working as I2C are presented in I2C section of chapter 15.

MSSP: Master Synchronous Serial Port - SPI: Serial Peripheral Interface: SCK SDI, SDO, SS

The MSSP: Master Synchronous Serial Port is a serial communication module that could be configured to communicate using I2C or SPI serial protocols. All the three PIC18

devices support SPI serial communication protocol. The PIC18 device could also be configured as a master or slave and can either transmit or receive the data and can generate clock or operate based on clock provided by external master. When MSSP is configured for SPI the serial clock is generated on SCK line. The SPI module of PIC18 may use the same SCK line when operated by clock from external master. The SPI protocol may be implemented in full duplex or half duplex mode. In full duplex mode the data transferred out of the module on SDO line and data received in on the SDI line. The SS function pin is used to select the slave by the master. For a slave it is input and output for a master. The details of the MSSP module working as SPI are presented in SPI section of chapter 15.

CAN: Controller Area Network Protocol: CANTX, CANRX

The CAN – Controller Area Network is another serial communication protocol that is based on BOSCH CAN 2.0 A/B specification protocol. The module supports CAN 1.2, CAN 2.0A, CAN 2.0B Passive and CAN 2.0B Active versions. Only the PIC18F458 has the CAN module. This module is provided with CANTX and CANRX function pins for data transmission and reception. The details of the CAN module are presented in CAN section of chapter 15.

Low Voltage Detect: LVDIN

All the three PIC18 devices consist of a Low Voltage Detect module. This is useful in determining if the minimum required voltage is reached upon start-up or if the voltage has dropped below a certain voltage level. An interrupt is generated if the required voltage level is not available. This is used to turn off the internal modules to avoid damage or incorrect operation when sufficient voltage level is not maintained.

The voltage level is software programmable and the bits LVDL3:LVDL0 of the LVDCON register selects a corresponding voltage step among the internal resistive ladder network. It may also be set to an external analog voltage applied through the LVDIN input function pin. The details of the same are presented in Low Voltage detect section of the special features in chapter 16.

ICSP: In-Circuit Serial Programming: PGM, PGC, PGD

The flash program memory is programmed using the ICSP programming function pins. The Low Voltage ICSP programming enable pin PGM must be set to 1 to enable the ICSP programming in low voltage. The PGM pin may be used as I/O when low voltage is not used. The low voltage ICSP programming is not available in PIC18C801. It is however present in PIC18F458 and PIC18F8720. The PGC and PGD act as the clock and data lines for the in-circuit serial programming.

EMI: External Memory Interface:

The three PIC18 devices presented in this book mainly vary in this particular aspect. One PIC18 doesn't contain ability to interface memory externally, one doesn't contain internal memory, all the required is external interfaced memory and the third has internal memory while also providing interface for external memory.

The PIC18F458 has a decent amount of inbuilt data memory – RAM & EEPROM, along with flash program memory. Hence it is not provided with external memory interface. The PIC18C801 is only provided with internal RAM data memory and does not contain EEPROM data memory or the Flash program memory. Hence it is mainly dependent on external interfaced flash memory. The PIC18F8720 has a decent amount inbuilt data memory – RAM & EEPROM, along with flash program memory. However, it provides the feature for interfacing additional memory externally through the external memory interface similar to PIC18C801. The memory capacities have been presented in the PIC18 introduction section and provided in detail later in chapter 8 which also presents Interfacing of external memory and the available external memory interface pins.

Since all the port pins and function/module wise pins are presented in the table 7.3 and 7.4, only the description of PIC18F458 PORT's & its functionalities is presented here. The additional special functionality that is present in the other two devices, not present in PIC18F458 is the external memory interface and it is discussed in detail later in chapter 8 related to memories. Interested readers may refer the respective data sheets of PIC18C801 & PIC18F8720 for any additional details or specific information related to a port or any of the functionalities.

7.2.1 Port A

This PORTA is a 7-bit bidirectional Port. It is a bit/byte addressable port. In addition to the I/O functionality (Input or Output) the pins of PORTA are used as analog inputs for the in-built 10-bit ADC, Comparator reference output voltage, supply reference voltage (V_{REF+} & V_{REF-}), external clock input for Timer0, Oscillator input, clock output, Slave select for SSP module and as a low voltage detect input. The details of specific multiplexed functionalities are presented in the PORTA functions of table 7.3.

The registers used for typical I/O functions of PORT A are TRISA, LATA & PORTA. To configure a port pin as input the corresponding TRISA bit must be set to 1 and must be cleared to 0 for output. When a pin of PORTA is configured as output, the value of the respective (LATA) output data latch bit is written to selected PORTA pin. When it is configured as input, the output driver is set to be in high impedance mode, and the input value is stored into the respective (LATA) input data latch bit.

Similar to other special/peripheral functionalities of a port, the special functions of PORTA override the general I/O function. Here they may be controlled through the respective special function control registers of ADC, reference voltage, Timer 0 Clock Input, Comparator module or SSP.

Upon a POR-Power on Reset, the RA0-RA3, RA5 are configured as analog inputs and are read as 0, whereas RA4 & RA6 are configured as digital inputs. The pins that are used as analog inputs for the ADC must be configured as inputs by setting the corresponding TRISA bits to 1 as the TRISA register controls the direction for the corresponding analog input pins of ADC module. Additionally, the configuration of ADC overrides the digital I/O feature with an analog input feature.

The RA4 pin is multiplexed with T0CKI and is Schmitt trigger input with an open drain output. The rest of PORTA pins have TTL input levels and full CMOS output drivers. All the I/O pins of port A have diode protection to V_{DD} and V_{SS} except for RA4 which has the diode protection only to V_{SS} .

7.2.2 Port B

This PORTB is an 8-bit bidirectional Port. It is a bit/byte addressable port. In addition to the I/O functionality (Input or Output) the pins of PORTB are used as External interrupts: INT0, INT1 & INT2, Interrupt on Change (on RB4-RB7), CANTX (transmit) & CANRX (receive) for the Controller Area Network – CAN's (Serial Communication protocol) and PGM (Enable), PGC (Clock) & PGD (Data) for the Serial Programming feature. The details of specific multiplexed functionalities are presented in PORTB functions of table 7.3.

The registers used for typical I/O functions of PORT B are TRISB, LATB & PORTB. To configure a port pin as input the corresponding TRISB bit must be set to 1 and must be cleared to 0 for output.

When a pin of PORTB is configured as output, the value of the respective (LATB) output data latch bit is written to selected PORTB pin. When it is configured as input, the output driver is set to be in high impedance mode, and the input value is stored into the respective (LATB) input data latch bit. The typical PIC18F458's structure of port B is shown in fig 7.4.

All the pins of port B are provided with a weak internal pull-up. An internal pull-up is helpful and is required when the device connected externally to the PORTB input pins is not capable to provide the sufficient voltage to maintain the required levels. The Pull-up is not required if the required levels could be maintained. The pull-up for all the pins of port B is enabled or disabled by a single control bit RBPU. The pull-up is enabled when the RBPU bit in INTCON2 register is cleared to 0 and is disabled when the control bit RBPU is set to 1.

When the port pins are configured as outputs or upon a POR, the weak pull-up is automatically disabled. Upon a POR the pull up is disabled by default as RBPU is set to 1.

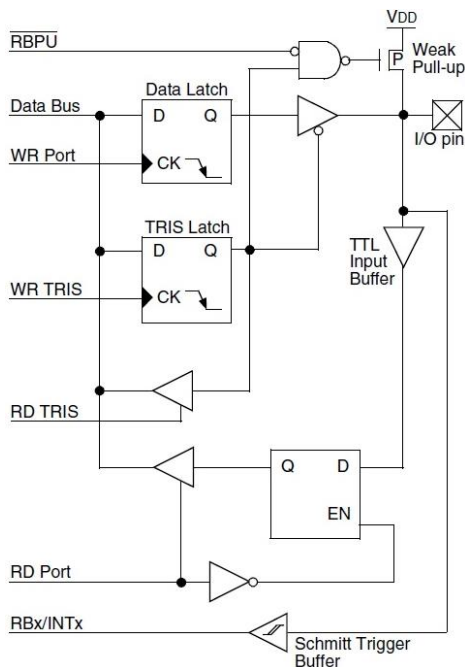


Fig 7.4 a) Pin 0, 1

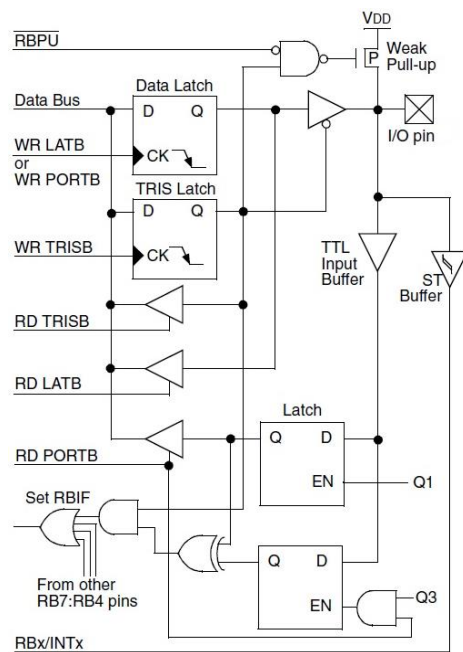


Fig 7.4 b) Pin 4, 5, 6 & 7

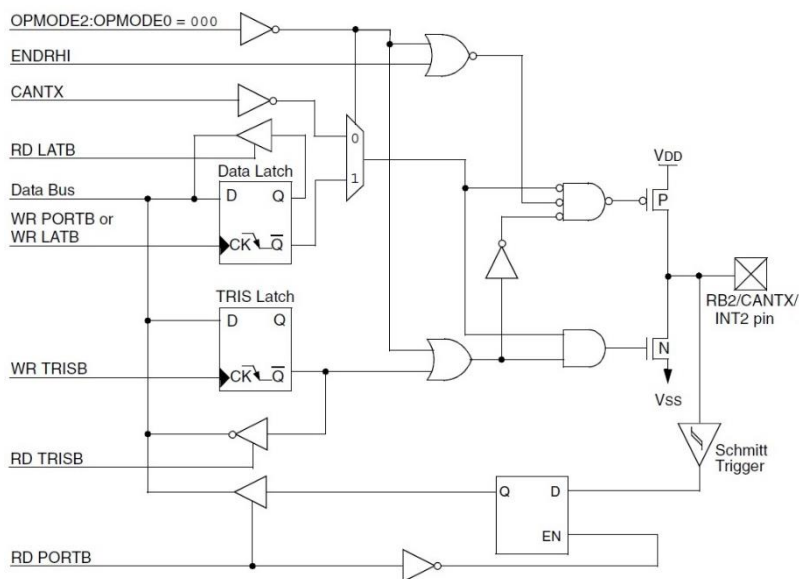


Fig 7.4 c) Pin 2

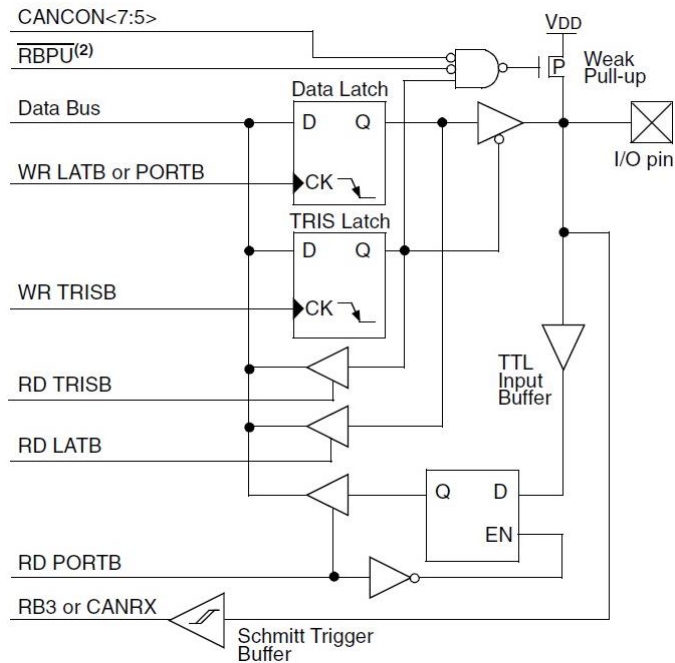


Fig 7.4 d) Pin 3

Fig 7.4 PIC18F458's Structure of Port B

Source: Microchip Document: PIC18FXX8 Data Sheet - DS41159E

The Interrupt on Change feature on RB4-RB7 and three external interrupts INT0, INT1, INT2 (and also INT3 in PIC18F8720) are available in PORTB. The interrupt on change feature of PORTB requires that the corresponding port pins must be configured as inputs by setting the corresponding TRISB bits to 1 to be able to generate the interrupt upon PORTB change. The TRISB register controls the direction for the corresponding interrupt on change feature. Additionally, reading PORTB may end a mismatch (interrupt on change) and clear the condition that sets the RBIF flag. Hence the LATB must be read instead of PORTB. The interrupt of change feature allows the microcontroller to wake-up the CPU from sleep. The Interrupt on Change feature and three external interrupts are discussed in detail in Interrupts section.

Similar to other special/peripheral functionalities of a port, the special functions of PORTB override the general I/O function. Here they may be controlled through the respective special function control registers of CAN, Interrupts or ICSP. As an example, the I/O pin functionality of RB5 is overridden and cannot be used when low voltage ICSP serial programming mode is used. The special features of PORTB are discussed detail in their respective modules.

In low voltage ICSP mode, the weak pull-up & I/O feature of RB5 pin is disabled. The pin must not be held low during normal operation to ensure against entry into ICSP mode. Additionally, when the pin is configured as output, the 5th bit of LATB must also be cleared to ensure correct operation.

Typically, all the port pins of PORTB have TTL input levels and CMOS output drivers. The port pins RB0-RB2 (RB3 in PIC18F8720) have Schmitt trigger levels when configured for external interrupts. Schmitt trigger levels are also maintained when configured in ICSP mode. All the I/O pins of port B have diode protection to V_{DD} and V_{SS} .

Operation of PIC18F458's Port B Pin 0 & 1:

The pin 0 and pin 1 of PORTB have identical structure and are used for external interrupts INTO & INT1. The operation and data flow for pin 0 and 1 of PORTB is shown fig 7.5 for a) Output, b) Input, c) Interrupt generation respectively.

Output Configuration:

The pin 0 of PORTB is configured as output when a logic 0 is written to TRIS<0> register. This is accomplished internally by writing logic 0 on data bus line 0 to configure as output and applying a negative edge signal: WR TRIS to the TRISB latch. When a pin is configured as output the internal weak pull-up is disabled. When a transfer of one – '1' to the I/O pin is desired, the corresponding value must be written into the corresponding PORTB<0> data latch. It is written internally by a negative edge write signal: WR PORT into the data latch. When the TRIS<0> bit is configured as output it enables the output driver to drive this data present in the data latch on to the I/O pin. The same applies for a transfer of logic zero – '0'. The application of write control signals and data transfer is shown as dotted line in fig 7.5a.

Input Configuration:

The pin 0 of PORTB is configured as input when a logic 1 is written to TRIS<0> register. This is accomplished internally by writing logic 1 on data bus line 1 to configure as input and applying a negative edge signal: WR TRIS to the TRISB latch. When a pin is configured as input the internal weak pull-up is enabled if internal weak pull-up is enabled by clearing the PORTB pull up enable bit - RBPU. In the case of an enabled internal pull-up the external input device must provide a ground path to send a logic 0 to PIC18's I/O pin. It is not required to drive the pin to VCC to send a logic 1 as the pull maintains the logic 1 level. In the case of disabled internal pull-up an external pull-up must be provided for driveability. The value of the I/O pin is read through the PORTB register. When PORTB is read, a RD port signal is internally generated to enable the input latch and input driver which places the I/O pin data on to the data bus. The application of read control signal and data transfer is shown as dotted line in fig 7.5b.

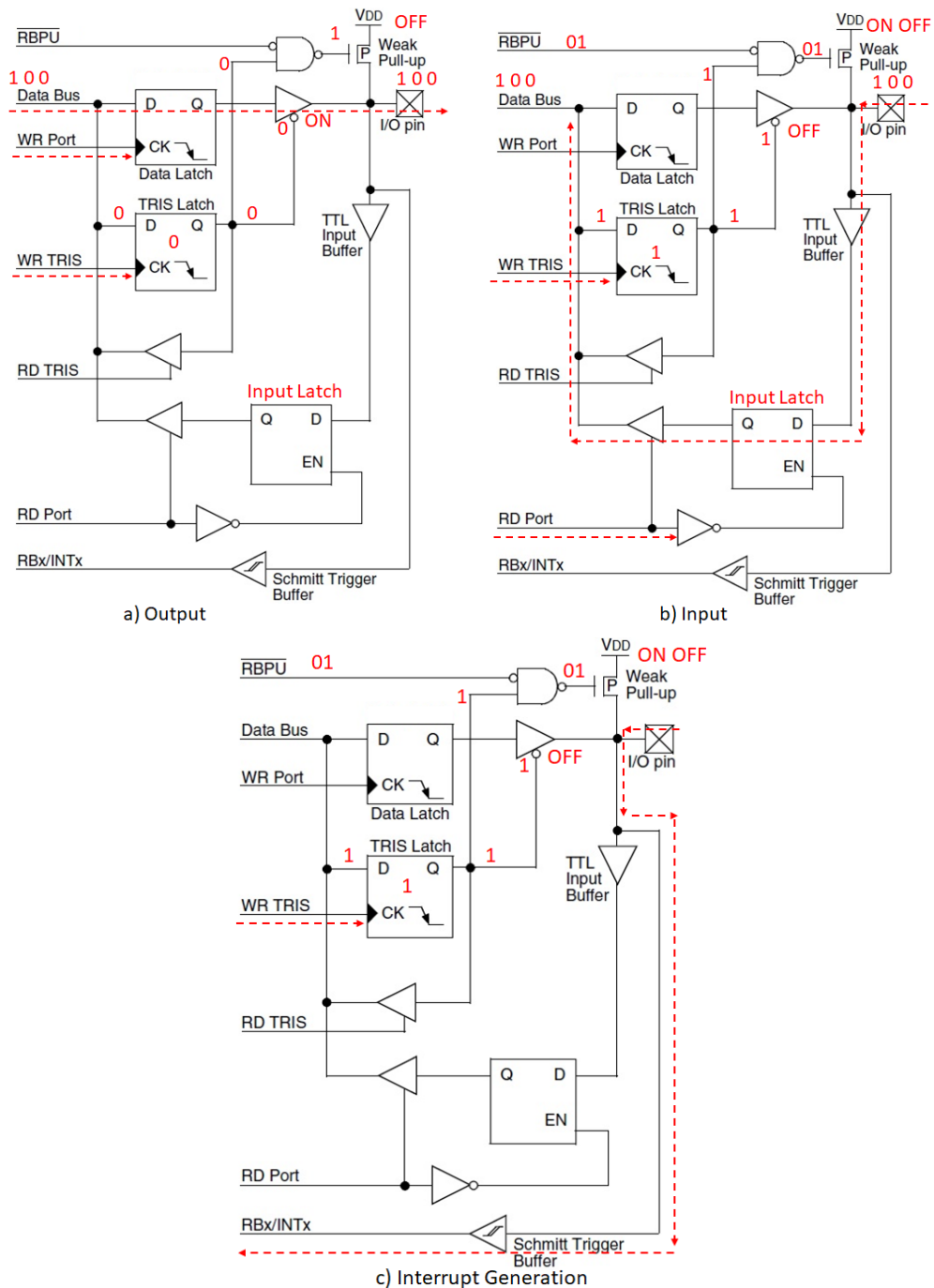


Fig 7.5 Operation of PIC18F458's Port B Pin 0 & 1

Source: Microchip Document: PIC18FXX8 Data Sheet - DS41159E

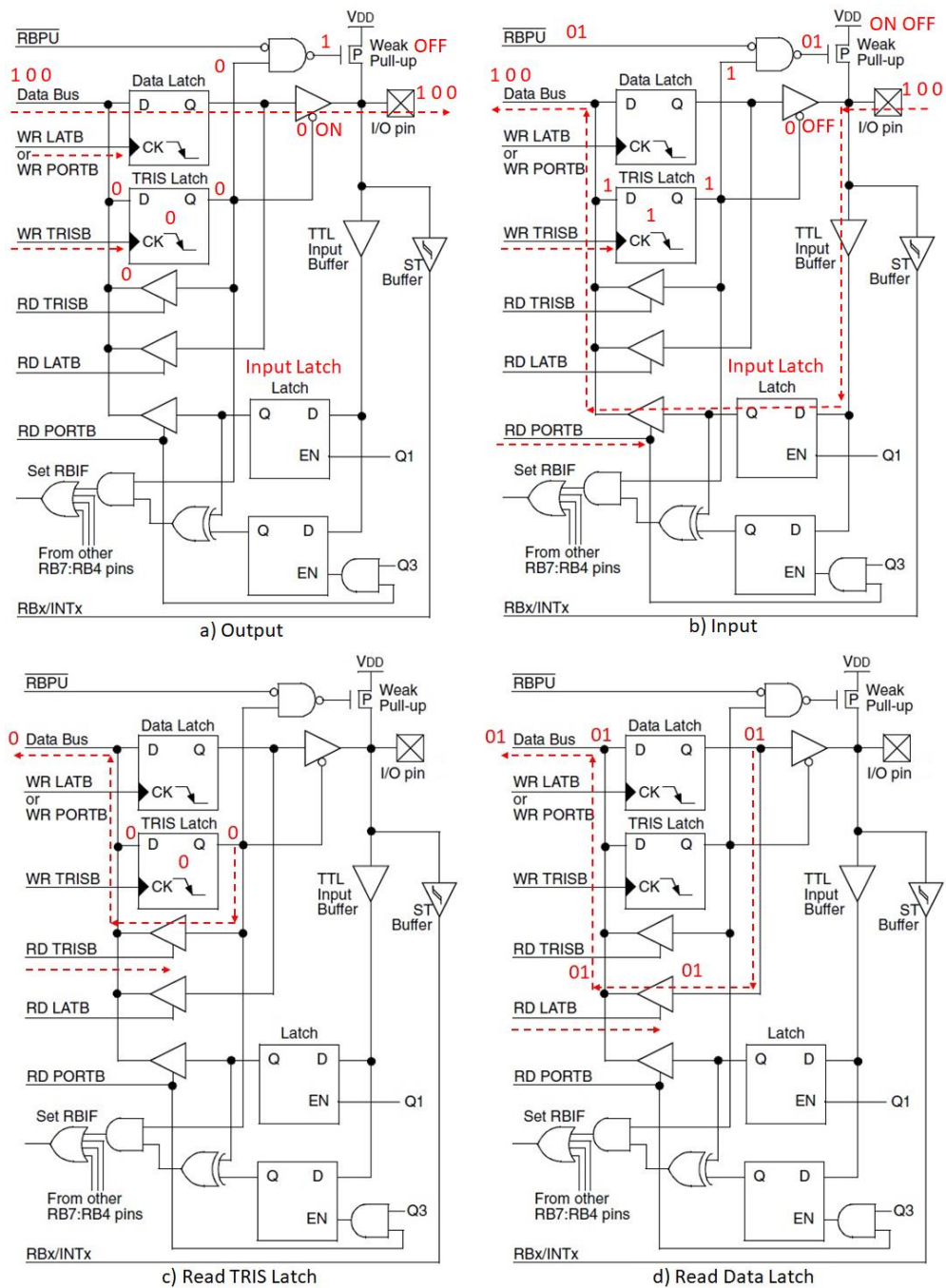


Fig 7.6 Operation of PIC18F458's Port B Pin 4 to 7

Source: Microchip Document: PIC18FXX8 Data Sheet - DS41159E

Operation of PIC18F458's Port B Pins 4 to 7:

The pins 4 to 7 of PORTB have identical structure and are used for PORTB Interrupt on change. The operation and data flow for pin 4 to 7 of PORTB is shown fig 7.6 for a) Output, b) Input, c) Reading TRIS Latch, d) Reading Data Latch respectively.

Read TRIS Latch:

The TRIS<4> register must be written with a 0 or 1 for configuring as output or input. The TRIS register is readable and writable. The value stored in the TRIS<4> latch is read when an internal read signal - RD TRISB is applied which enables the driver to place the data of the TRIS latch on to the data bus. The application of read control signal and data transfer is shown as dotted line in fig 7.6c.

Read Data Latch:

When a PORTB<4> is written with a 1 or 0, the value is written to the Data Latch<4>. When the TRIS<4> bit is set to 0, the pin is configured as output and the value in the latch is driven to the PORT<4> I/O pin. When a read operation on PORTB<4> is performed the port I/O pin is read. When the LATB<4> register is read, the Data latch<4> is read when a latch read signal - RD LATB is applied which enables the driver to place the data of the data latch on to the data bus. The application of read control signal and data transfer is shown as dotted line in fig 7.6d. The Read-Modify-Write operations on LAT registers, read and write latched output value for PORT. When a file register is used as part of any instruction, a Read-Modify-Write operation is performed. A write operation also performs a read operation.

7.2.3 Port C

This PORTC is an 8-bit bidirectional Port. It is a bit/byte addressable port. In addition to the I/O functionality (Input or Output) the pins of PORTC are used as Timer 1 Oscillator: T1OSO & T1OSI, Timer 1 Clock Input: T1CKI, CCP – Capture Compare output: CCP1, SSP-Synchronous Serial Port (I2C/SPI): Serial Clock SCL/SCK, Serial Data SDA/SDI and USART-Universal Synchronous Asynchronous Receive Transmit's: Transmit TX, Clock CK, Receive Rx and Data DT features. The details of specific multiplexed functionalities are presented in PORTC functions of table 7.3.

The registers used for typical I/O functions of PORTC are TRISC, LATC & PORTC. To configure a port pin as input the corresponding TRISC bit must be set to 1 and must be cleared to 0 for output. When a pin of PORTC is configured as output, the value of the respective (LATC) output data latch bit is written to selected PORTC pin. When it is configured as input, the output driver is set to be in high impedance mode, and the input value is stored into the respective (LATC) input data latch bit.

Similar to any other special/peripheral functionalities of a port, the special functions of PORTC override the general I/O function. Here they may be controlled through the respective special function control registers of Timer 1, SSP or USART. As an example, the I/O pin functionality of RC0 and RC1 is overridden with timer oscillator inputs. They cannot be used as I/O when Timer1 oscillator is enabled and the TRISC bit<1:0> value is overridden and ignored. The special features of PORTC are discussed in detail in their respective modules. All the port pins of PORTC have Schmitt trigger levels. All the I/O pins of port C have diode protection to V_{DD} and V_{SS} .

7.2.4 Port D

This PORTD is an 8-bit bidirectional Port. It is a bit/byte addressable port. In addition to the I/O functionality (Input or Output) the pins of PORTD are used for PSP- Parallel Slave port as parallel data lines PSP0:PSP7, Analog Comparator inputs C1IN+, C1IN-, C2IN+ and C2IN-, ECCP1 – Enhanced Capture Compare output and PWM features. The details of specific multiplexed functionalities are presented in PORTD functions of table 7.3.

The registers used for typical I/O functions of PORTD are TRISD, LATD & PORTD. To configure a port pin as input the corresponding TRISD bit must be set to 1 and must be cleared to 0 for output. When a pin of PORTD is configured as output, the value of the respective (LATD) output data latch bit is written to selected PORTD pin. When it is configured as input, the output driver is set to be in high impedance mode, and the input value is stored into the respective (LATD) input data latch bit.

Similar to any other special/peripheral functionalities of a port, the special functions of PORTD override the general I/O function. Here they may be controlled through the respective special function control registers of PSP, analog Comparator or ECCP. As an example, the I/O pin functionality of RD0 to RD7 is overridden with PSP mode. They cannot be used as I/O when PSP is enabled and the TRISD bit<7:0> value is overridden and ignored. The special features of PORTD are discussed in detail in their respective modules. Typically, all the port pins of PORTD have Schmitt Trigger levels in I/O mode and TTL levels when in PSP mode. All the I/O pins of port D have diode protection to V_{DD} and V_{SS} .

7.2.5 Port E

This PORTE is a 3-bit bidirectional Port. It is a bit/byte addressable port. In addition to the I/O functionality (Input or Output) the pins of PORTE are used for PSP – Parallel Slave Port control signals Read: RD, Write: WR, Chip Select CS and analog comparator 1 & 2 output and ADC's analog input AN5:AN7 features. The details of specific multiplexed functionalities are presented in PORTE functions of table 7.3.

The registers used for typical I/O functions of PORT E are TRISE, LATE & PORTE.

Even though the port E is a 3-bit port, the TRISE register is 8-bit and contains the direction control bits along with control bit for enabling the PSP mode and the respective PSP status bits that indicate the input buffer full, output buffer full and input buffer overflow detection bits.

To configure a port pin as input the corresponding TRISE bit must be set to 1 and must be cleared to 0 for output. When a pin of PORTE is configured as output, the value of the respective (LATE) output data latch bit is written to selected PORTE pin. When it is configured as input, the output driver is set to be in high impedance mode, and the input value is stored into the respective (LATE) input data latch bit.

Similar to any other special/peripheral functionalities of a port, the special functions of PORTE override the general I/O function. Here they may be controlled through the respective special function control registers of ADC, PSP or analog Comparator. As an example, the I/O pin functionality of RE0 to RE2 is overridden with PSP mode and RE1, RE2 are overridden with analog comparators 1 & 2's outputs. The RE0:RE2 pins cannot be used as I/O when PSP is enabled and the TRISE bit<2:0> value is overridden and ignored.

The configuration of ADC overrides the digital I/O feature with an analog input feature. Additionally, the pins that are used as analog inputs for the ADC must be configured as inputs by setting the corresponding TRISE bits to 1 as the TRISE register controls the direction for the corresponding analog input pins of ADC module.

The special features of PORTE are discussed in detail in their respective modules. Typically, all the port pins of PORTE have Schmitt Trigger levels in I/O mode and TTL levels when in PSP mode. All the I/O pins of port E have diode protection to V_{DD} and V_{SS} .

7.3 Interfacing & Programming of LED's

Light Emitting Diode – LED are class of diodes that emit light. They can be used as actuators. Similar to a diode these are directional and emit light when they are placed in circuit in the right direction.

Also similar to a diode they have a forward voltage drop V_f . This is typically 0.7V for a diode and 1.5V to 3V for an LED. A filament in a light bulb burns out over usage and when a current is applied beyond its specifications. Similarly, the LED Diodes have typical current limit I_d of around 20mA beyond which the LED burns out. So, a current limiting resistor must be connected in series with the microcontroller and LED as shown in Fig 7.7. The size of the limiting resistor is calculated as follows.

If the forward voltage drop V_f is 2V, and available voltage V_{CC} at the pin of the microcontroller is 5V then the remaining voltage is $V_e = 5 - 2 = 3V$. For a current limit of 20mA the resistance is calculated as $R = V_e / I_d = (V_{CC} - V_f) / I = (3V) / (20mA) = 150 \text{ Ohms}$. The current limit may vary based on the diode and usage. Hence a resistance of around 220Ω to 470Ω is typically used.

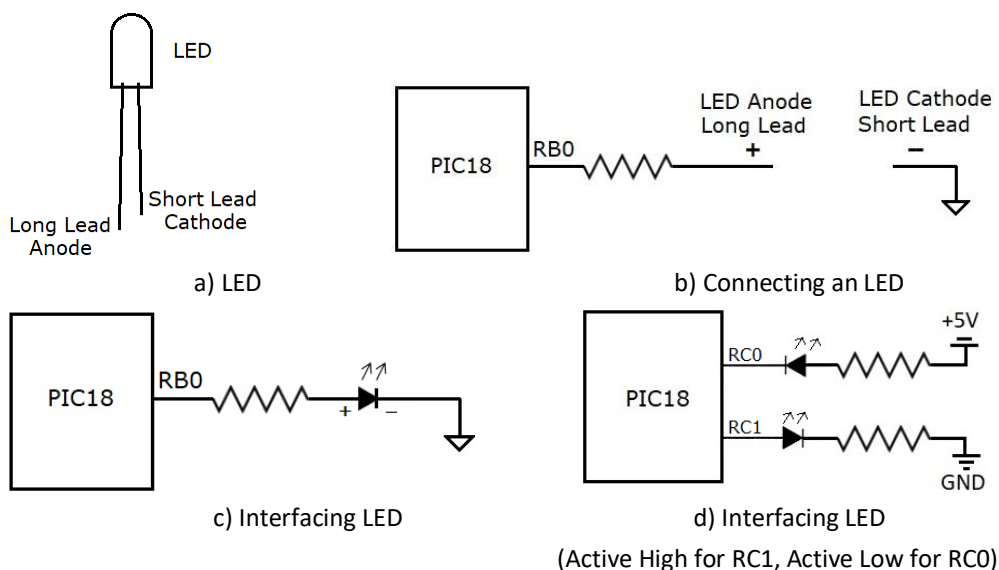


Fig 7.7 LED & its interfacing

A representation of LED is shown in fig 7.7a and LED connection is shown in fig 7.7b. LEDs may be interfaced in two ways: 1) Active High logic & 2) Active Low logic. The LED interfaced using active high logic as shown in fig 7.7c. The fig 7.7d shows the active high logic LED connection to RC1 whereas connection to RC0 shows the active low logic.

An active high logic requires a port pin to be set for the LED to be turned ON and port pin to be cleared for it to be turned OFF. The active low logic requires a port pin to be cleared for the LED to be turned ON and port pin to be set for it to be turned OFF.

For an LED connected in active high logic to RB0 of fig 7.7c or RC1 as shown in fig 7.7d, setting the port pin using instruction “BSF f, b, a” provides a +5V at the pin which allows the passage of current through the LED and turns the LED ON. Clearing the port pin using instruction “BCF f, b, a” provides a 0V at the pin and no current flows through the LED causing the LED to be turned OFF. For an LED connected in active low logic to RC0 as shown in fig 7.7d, setting the port pin using instruction “BSF f, b, a” provides a +5V at the pin and allows no current to flow through the LED causing the LED to be turned OFF. Clearing the port pin using instruction “BCF f, b, a” provides a 0V at the pin which allows the passage of current through the LED and turns the LED ON.

Program 7.1: Blink an LED connected at Port B pin 0.

The following program is used to demonstrate fig P7.1a as well as fig P7.1b. The pull-ups of port B are disabled when configured as output and the initial value of PORTB is 0. Hence for circuit designed as shown in fig P7.1a the first toggle generates a +5V at RB0 and causes the LED to be ON and the second toggle generates a 0V and causes the LED to be OFF.

For a circuit designed as shown in fig P7.1b the first toggle generates a +5V at RB0 and causes the LED to be OFF and the second toggle generates a 0V and causes the LED to be ON.

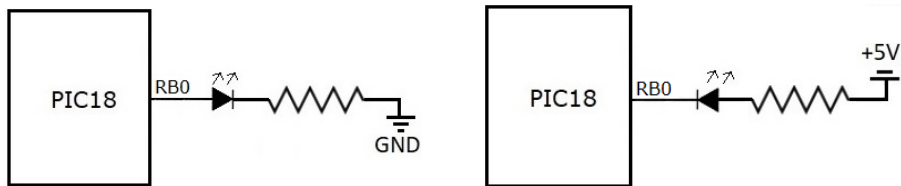


Fig P7.1 LED its interfacing in a) Active High Logic & b) Active Low Logic

	#INCLUDE <P18F458.INC>; CAN BE OTHER PROCESSOR		
	ORG	0x00	; Set Origin 0x0000 of Flash memory
	GOTO	START	; Go to Start of Main program.
START	ORG	0x100	; Set Origin 0x100 for Main program
	BCF	TRISB,0	; TRISB.0 = 0, Make PORTB Pin 0 as Output
L1:	BTG	PORTB,0,0	; Toggle PORTB Pin 0, PORTB.0 = PORTB.0
	CALL	DLY	; Call Module DLY to create a delay
	BRA	L1	; Branch to L1 to repeat change state/blinking
	ORG	0x200	; Set Origin 0x200
DLY	MOVLW	0x40	; WREG = 0x40
	MOVWF	0x10,0	; [0x10] = WREG = 0x40
DL1:	DECF	0x10,F	; Decrement File Register 0x10, [0x10] = [0x10] - 1
	BNZ	DL1	; If not zero Branch to DL1.
	RETURN		; Return from Delay module
	END		

Program 7.2: Blink an LED connected at Port B pin 0 with unequal LED ON time (Delay1) and LED OFF time (Delay2).

The following program could use the design arrangement of fig P7.1a or fig P7.1b. The Delay 1 is used as LED ON time and Delay 2 as LED OFF time for the circuit designed as shown in fig P7.1a. For the circuit designed as shown in fig P7.1b delays or the instructions “BSF PORTB,0” and “BCF PORTB,0” may be interchanged to achieve the same functionality.

The calculation for exact amount of delay that is created may be done similar to the example discussed in program 9.1 and 9.2 of the timer’s section. A program with generic delays is given below.

	#INCLUDE <P18F458.INC>; CAN BE OTHER PROCESSOR		
	ORG	0x00	; Set Origin 0x0000 of Flash memory
	GOTO	START	; Go to Start of Main program.

START	ORG	0x100	; Set Origin 0x100 for Main program
	BCF	TRISB,0	; TRISB.0 = 0, Make PORTB Pin 0 as Output
L1:	BSF	PORTB,0	; Set PORTB Pin 0, PORTB.0 = 1
	CALL	DLY1	; Call Module DLY1 to create a delay
	BCF	PORTB,0	; Clear PORTB Pin 0, PORTB.0 = 0
	CALL	DLY2	; Call Module DLY2 to create a delay
	BRA	L1	; Branch to L1 to repeat change state/blinking
	ORG	0x200	; Set Origin 0x200
DLY1	MOVLW	0x40	; WREG = 0x40
	MOVWF	0x10,0	; [0x10] = WREG = 0x40
DL1:	DECF	0x10,F	; Decrement File Register 0x10, [0x10] = [0x10] - 1
	BNZ	DL1	; If not zero Branch to DL1.
	RETURN		; Return from Delay module
	ORG	0x300	; Set Origin 0x300
DLY2	MOVLW	0x30	; WREG = 0x30
	MOVWF	0x10,0	; [0x10] = WREG = 0x30
DL3:	DECF	0x10,F	; Decrement File Register 0x10, [0x10] = [0x10] - 1
	BNZ	DL3	; If not zero Branch to DL3.
	RETURN		; Return from Delay module
	END		

Program 7.3: A set of 8 LEDs are connected to port B. Turn ON the LEDs connected to lower nibble even bits of Port B by an amount of delay 1 and Turn ON the LEDs connected to higher nibble odd bits of Port B by an amount of delay 2 alternately.

This implies that the ON time of lower nibble Even LEDs = delay1 and OFF time = delay2, the ON time of higher nibble Odd LEDs = delay2 and OFF time = delay1. The byte format is given below.

	High Nibble				Low Nibble			
EVEN/ ODD	O	E	O	E	O	E	O	E
BIT No	7	6	5	4	3	2	1	0
Value	0	0	0	0	0	1	0	1
HEX Value	0x05							
EVEN/ ODD	O	E	O	E	O	E	O	E
BIT No	7	6	5	4	3	2	1	0
Value	1	0	1	0	0	0	0	0
HEX Value	0xA0							

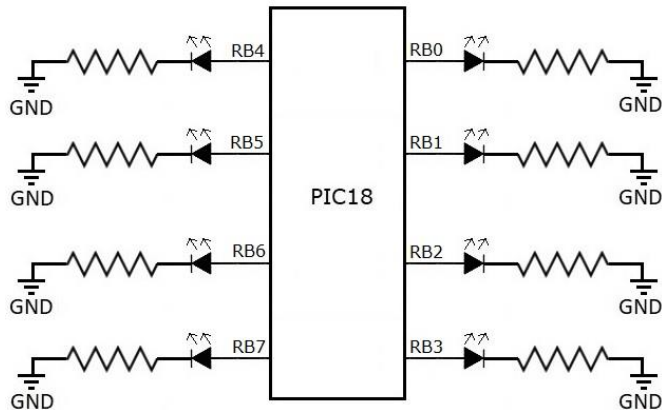


Fig P7.3 Interfacing 8 LEDs to Port B in active high logic

The interfacing circuit is shown in fig P7.3 and the program is as follows.

	#INCLUDE <P18F458.INC>; CAN BE OTHER PROCESSOR		
	ORG	0x00	; Set Origin 0x0000 of Flash memory
	GOTO	START	; Go to Start of Main program.
START	ORG	0x100	; Set Origin 0x100 for Main program
	MOVLW	0x00	; WREG = 0x00
	MOVWF	TRISB,0	; TRISB = 0x00, Make PORTB as Output
L1:	MOVLW	0x05	; WREG = 0x05
	MOVWF	PORTB,0	; PORTB = 0x05, Turn ON the low nibble even LEDs
	CALL	DLY1	; Call Module DLY1 to create a delay1
	MOVLW	0xA0	; WREG = 0xA0
	MOVWF	PORTB,0	; PORTB = 0xA0, Turn ON the high nibble odd LEDs
	CALL	DLY2	; Call Module DLY2 to create a delay2
	BRA	L1	; Branch to L1 to repeat change state/blinking
	ORG	0x200	; Set Origin 0x200
DLY1	MOVLW	0x40	; WREG = 0x40
	MOVWF	0x10,0	; [0x10] = WREG = 0x40
DL1:	DECF	0x10,F	; Decrement File Register 0x10, [0x10] = [0x10] - 1
	BNZ	DL1	; If not zero Branch to DL1.
	RETURN		; Return from Delay module
	ORG	0x300	; Set Origin 0x300
DLY2	MOVLW	0x30	; WREG = 0x30
	MOVWF	0x10,0	; [0x10] = WREG = 0x30
DL3:	DECF	0x10,F	; Decrement File Register 0x10, [0x10] = [0x10] - 1
	BNZ	DL3	; If not zero Branch to DL3.
	RETURN		; Return from Delay module

7.4 Interfacing & Programming of Buttons/Switches

One of the common inputs for a computer is the keyboard which consists of multiple keys that are used as a group. Similarly, to keys are the buttons & switches that are commonly used as individual elements.

A sample image of button, switch is shown in fig 7.8a & 7.8b, taken from sparkfun website. There are two basic types of button or switches: 1) Normally Open – NO, 2) Normally Close – NC type. A representation of the same is shown in fig 7.8c and fig 7.8d respectively. The pushing of button is shown in fig 7.8e and fig 7.8f which represent a normally open switch before a push/press and when it is pushed/pressed.

The switch has two terminals and is open on both ends where one end may be connected to a microcontroller and the other end may be connected to VDD (+5V) or VSS (GND). Similar to LED's an active high logic or an active low logic may be formed with buttons/switches. The switch connection to RB5 in fig 7.9a forms an active high logic and the connection of RB6 forms the active low logic. An active low logic here requires an internal pull up in microcontroller's port pins or an external pull-up on the port pin outside the IC chip. An active high logic may include an external pull down outside of the IC chip to ensure that the terminal is grounded. The pull-up and pull down would make sure that the pin is not floating. These are shown in the practical switch interfacing circuit of fig 7.9b.

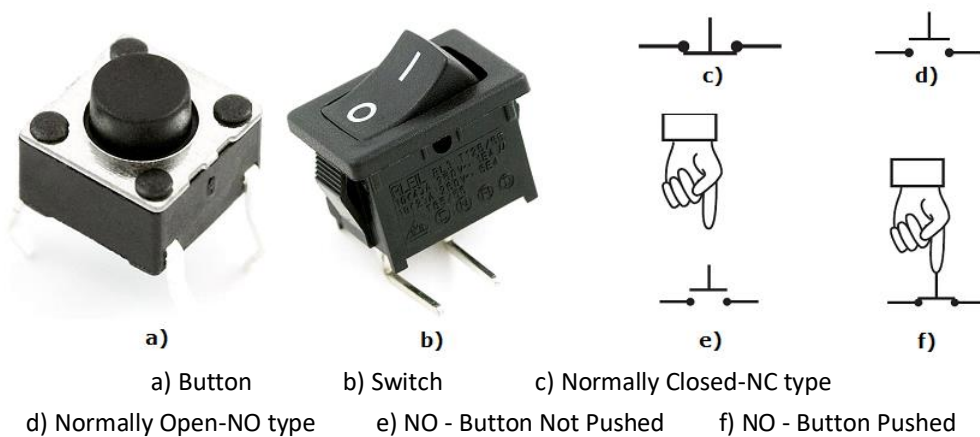


Fig 7.8 Button/Switch & its representations from Sparkfun

For the active high logic setup of RB5 pin in fig 7.9b, when the button is not pressed the port pin RB5 presents logic 0 when the port is read. This is because the switch is open and port pin is isolated from the VDD of +5V and it is connected to a pull-down resistor R_D . This pull-down resistor ensures logic 0 level at the port. When an external pull-down like this

is provided internal pull-down is not required, but since PIC18 does not have an internal programmable pull down, an external pull-up is recommended.

When the button is pressed the port pin RB5 presents logic 1 when the port is read. This is because the switch gets closed when the button is pressed and hence connects to VDD of +5V providing a logic 1.

For the active low logic setup of RB6 pin in fig 7.9b, when the button is not pressed the port pin RB6 presents logic 1 when the port is read. This is because the switch is open and port pin is isolated from the VSS/GND and it is connected to a pull up resistor R_U . This pull-up resistor ensures logic 1 level at the port. When an external pull-up like this is provided internal pull-up is not required and vice-versa.

When the button is pressed the port pin RB6 presents logic 0 when the port is read. This is because the switch gets closed when the button is pressed and hence connects to VSS/GND providing logic 0.

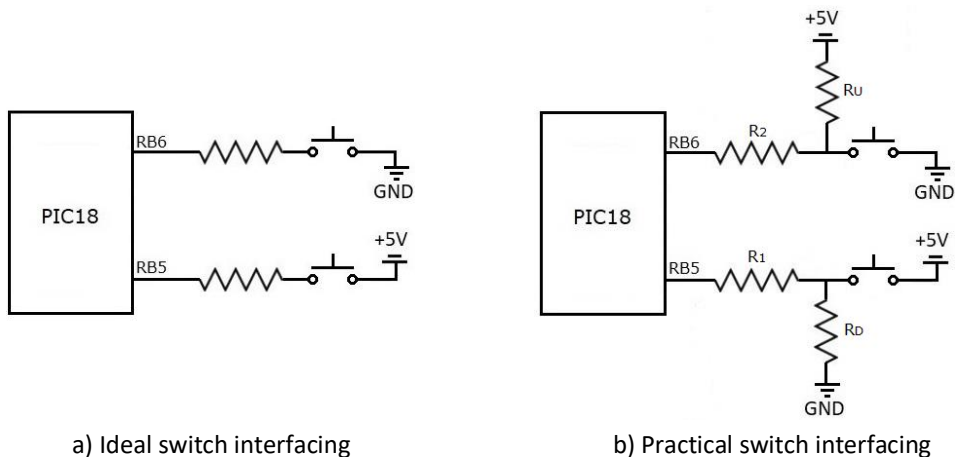


Fig 7.9 Button/Switch & its Interfacing

Program 7.4: A button/switch is connected to the pin 5 of PORTB and an LED to pin 0 of PORTC. Read the status of the button/switch and blink the LED if the switch is pressed.

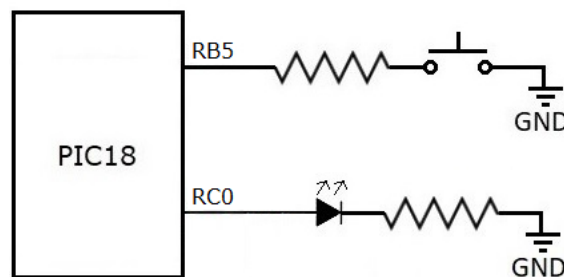


Fig P7.4 Button/Switch & its LED Interfacing

	#INCLUDE <P18F458.INC>; CAN BE OTHER PROCESSOR		
	ORG	0x00	; Set Origin 0x0000 of Flash memory
	GOTO	START	; Go to Start of Main program.
	ORG	0x100	; Set Origin 0x100 for BC_LS
BC_LS	MOVF	PORTB,0	; Read PORTB to WREG, WREG = PORTB
	ANDLW	0x20	; Extract status of Port B pin 5
	BZ	IL1	; If RB5 is 0, go to IL1
	BCF	PORTC,0	; RB5 is 1, Button Not Pressed ; So turn OFF the LED at RC0
	BRA	IL2	; Go to IL2
IL1	BSF	PORTC,0	; RB5 is 0, Button Pressed ; So turn ON the LED at RC0
IL2	RETURN		; Return from BC_LS module
	ORG	0x200	; Set Origin 0x200 for Main program
START	BSF	TRISB,5	; TRISB.5 = 1, Make PORTB Pin 5 as Input
	BCF	TRISC,0	; TRISC.0 = 0, Make PORTC pin 0 as output
	BCF	INTCON2,RBPU	; Enable the weak pull up for PORTB ; Internal weak Pull-up is enabled,
M1	CALL	BC_LS	; Call BC_LS module
	BRA	M1	; Branch to M1 to repeat change state/blinking
	END		

This program makes use of weak pull-up provided by PORTB. Hence external pull-up is not required and not used. It uses an active low logic for button and active high logic for LED. The setup is shown in fig P7.4.

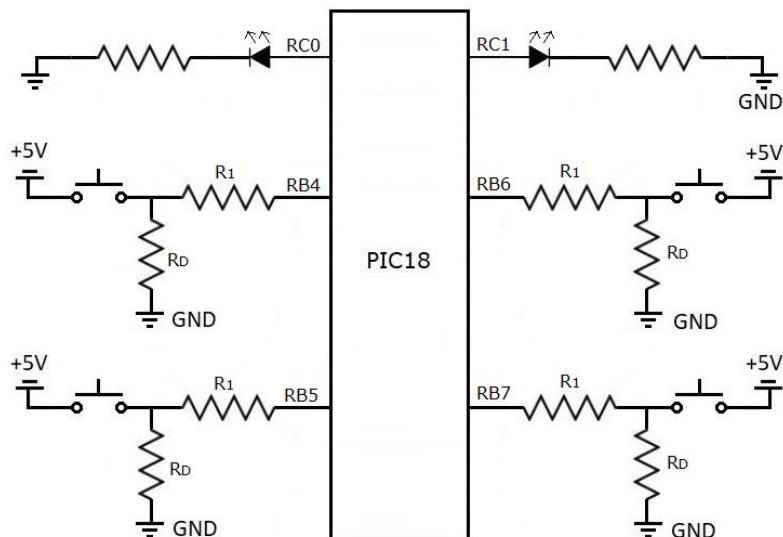


Fig P7.5 Button/Switch & its LED Interfacing

Program 7.5: Four buttons/switches are connected to the pin 4, 5, 6 & 7 of PORTB and two LEDs to pin 0 and 1 of PORTC. The button at pin 4 is denoted on the LEDs as 00, 5 as 01, 6 as 10 and 7 as 11, where 1 represents LED ON and 0 represents an LED OFF. Use polling to obtain the status of the buttons/switches and indicate the button number by turning on the correct combination of LEDs. Here it is assumed that no two switches are pressed at the same time.

	#INCLUDE <P18F458.INC>; CAN BE OTHER PROCESSOR		
	ORG	0x00	; Set Origin 0x0000 of Flash memory
	GOTO	START	; Go to Start of Main program.
	ORG	0x100	; Set Origin 0x100 for RB ISR
BC_LS	MOVF	PORTB,0	; Read PORTB to WREG, WREG = PORTB
	ANDLW	0xF0,0	; Extract status of switches, RB4-RB7
	BTFSC	WREG,4	; Check if switch 4 is pressed
	GOTO	L1	; Go to L1
	BTFSC	WREG,5	; Check if switch 5 is pressed
	GOTO	L2	; Go to L2
	BTFSC	WREG,6	; Check if switch 6 is pressed
	GOTO	L3	; Go to L3
	BTFSC	WREG,7	; Check if switch 7 is pressed
	GOTO	L4	; Go to L4
	RETURN		; Return from BC_LS module
L1	BCF	PORTC,0	; Set combination for Button at Pin 4, LED 0 OFF
	BCF	PORTC,1	; Set combination for Button at Pin 4, LED 1 OFF
	RETURN		; Return from BC_LS module
L2	BSF	PORTC,0	; Set combination for Button at Pin 5, LED 0 ON
	BCF	PORTC,1	; Set combination for Button at Pin 5, LED 1 OFF
	RETURN		; Return from BC_LS module
L3	BCF	PORTC,0	; Set combination for Button at Pin 6, LED 0 OFF
	BSF	PORTC,1	; Set combination for Button at Pin 6, LED 1 ON
	RETURN		; Return from BC_LS module
L4	BSF	PORTC,0	; Set combination for Button at Pin 7, LED 0 ON
	BSF	PORTC,1	; Set combination for Button at Pin 7, LED 1 ON
	RETURN		; Return from BC_LS module
	ORG	0x200	; Set Origin 0x200 for Main program
START	SETF	TRISB,0	; Set all the pins of PORT B as inputs, TRISB = 0xFF
	CLRF	TRISC,0	; Set all the pins of PORT C as outputs, TRISC = 0x00
M1	CALL	BC_LS	; Call BC_LS module
	BRA	M1	; Branch to M1 to repeat change state/blinking
	END		

This program does not use internal pull-up provided by PORTB but uses an external pull-down with active high logic for buttons & LEDs. The setup is shown in fig P7.5.

7.5 Overview of Interrupts & its Structure

The concept of Polling, Interrupts and its structure in PIC18 devices is presented in this section.

Polling

Polling is the process of checking for a set/reset condition through instructions in the user's software program. Checking a specific modules flag bit, i.e. a timer flag TMR1IF using instructions repetitively as shown below is an example for polling.

L1	BTFSS	T1CON, TMR1IF	(Or)	L1	BTFSS	T1CON, TMR1IF
	BRA	L1			BRA	L2
					BRA	Req_Op
				L2	NOP	; Any other alternate required logic
					BRA	L1
						Req_Op

Here, the above instructions must be continuously executed by the CPU to determine if the timer flag is set. The CPU is kept busy checking, if the flag is set and cannot perform any other task, bringing the efficiency down, increasing the execution time.

Interrupt

In such a case as shown above, a technique that may be employed to improve the CPUs efficiency is the use of interrupts. In Embedded systems terminology, Interrupt is an important signal generated by one or more of the peripherals that are either internal or external to the microcontroller. These signals are detected by the hardware instead of the user's software instructions and are used to interrupt the current execution sequence to perform a high priority task related to that interrupt.

A Fire alert/climate control system in houses or industries serves a good example to demonstrate the need and applicability of interrupts. Such a system uses one or more of fire or smoke sensors that are interfaced to a microcontroller. The microcontroller may be interfaced to water sprinkler system and possibly working also as a climate/temperature control system. Now the regular or main function of this system is to regulate the temperature inside the house or industry to maintain healthy working environment and to interface fire sensor signals to detect using microcontroller & to turn ON the sprinkler system in case of a fire.

Now to perform this task separate microcontrollers may be used to continuously check the existence of a fire and temperature, or a single microcontroller device may be used for all the functions through use of interrupts. A single system setup is typically designed to

continuously monitor temperature without interrupt and interface fire sensor signal as an interrupt to detect the fire. Upon the detection of fire through the interrupt signal, the sprinklers may be turned ON through the interrupt service routine – ISR. The example that is presented in not highly time critical but serves the basic purpose to demonstrate the use of interrupt. An example for highly time critical or real time system may be involved in aircraft control system, industrial plants etc.

Interrupts are controlled: enabled or disabled at two levels, i.e. Global level and local or individual level. At the global level they are operated and controlled through three bits: GIE, GIEL/PEIE and IPEN. At the local/individual level they are operated and controlled through another three bits: xxIF, xxIE & xxIP bits. Here ‘xx’ corresponds to the related module/functionality present in the PIC18 microcontroller. The functionality of all the 6 bits is presented here.

Global Level Interrupt control bits

- Global Interrupt Enable bit – GIE, controls all the interrupts. This acts as the master control. Interrupts are globally enabled when GIE bit is set to 1 and disabled when this bit is set to 0. This bit is accessible through the Interrupt Control - INTCON register.
- Interrupt Priority Enable bit – IPEN, enables or disables the priority levels. When this bit is set to 1, the priority levels PIC18 are enabled and activates two priority levels with vector location 0x08 for high priority and 0x18 for low priority. When this bit is cleared, priority levels are disabled and it is said to be in PIC16 compatibility mode (PICmicro mid-range) where there is only one priority level – high priority with vector location 0x08. In single priority mode the individual priority bits do not have any effect. This bit is accessible through the Reset Control – RCON register.
- Peripheral Interrupt Enable bit - GIEL/PEIE, acts as an additional global control for low priority interrupts. This controls all the interrupt sources configured for low priority when priority levels are enabled. It is not required for high priority interrupts. This bit is accessible through the Interrupt Control - INTCON register.

Summary of Global Interrupts:

GIE/GIEH: Global Interrupt Enable bit

When IPEN (RCON<7>) = 0

1 = Enables all unmasked interrupts

0 = Disables all interrupts

When IPEN (RCON<7>) = 1

1 = Enables all high priority interrupts

0 = Disables all priority interrupts

PEIE/GIEL: Peripheral Interrupt Enable bit

When IPEN (RCON<7>) = 0

1 = Enables all unmasked peripheral interrupts

0 = Disables all peripheral interrupts

When IPEN (RCON<7>) = 1

1 = Enables all low priority peripheral interrupts
0 = Disables all low priority peripheral interrupts

Local or Individual Level Interrupt control bits

- Flag bit - xxIF is a status flag bit that indicates occurrence of interrupt event, ex: Timer1 overflow flag - TMR1IF bit.
- Enable bit – xxIE is a control bit that enables or disables the generation of an interrupt, ex: Timer1 overflow enable - TMR1IE bit.
- Priority bit – xxIP is a control bit that sets the interrupt source as a high or low priority, ex: Timer1 overflow interrupt priority - TMR1IP bit.

There are multiple modules & sources in PIC18 and hence have multiple local interrupt control bits. A complete list of all modules, its control flags, associated registers for the different devices are presented in table 7.5 through to table 7.12.

Requirements for Generation of an Interrupt

When the required conditions for an interrupt are met, an interrupt would be generated. The common PIC18 requirements for generation of interrupts are

- 1) Corresponding individual interrupt flag must be set, i.e. xxIF bit set
- 2) The specific Interrupt is enabled, i.e. xxIE bit is set
- 3) Global Interrupt is enabled, i.e. GIE bit is set
- 4) Corresponding priority bit is set/reset if priority levels (IPEN is set) are enabled, i.e. xxIP is set/reset
- 5) Peripheral Interrupt is enabled, i.e. PEIE/GIEL is set if required (required for interrupts configured as low priority).

The individual interrupt flag - xxIF flag may be set/reset irrespective of the status of the individual interrupt enable bit, the global interrupt enables bits or the priority level bit. It is set based on the individual module's functionality. It can be reset using instructions by the user's software.

Classification of Interrupts

Interrupts may be classified in different ways.

- Hardware vs Software
- Internal vs External
- Maskable vs Nonmaskable
- Hardware module-based classification

Hardware Interrupts

Interrupts are termed as hardware interrupts if they are purely generated due to a specific hardware component, internal or external to the microcontroller. All the interrupts including a Reset could be generated when respective conditions are set by using hardware. Hence all interrupts in PIC18 devices may be considered as hardware interrupts.

Software Interrupts

Interrupts are termed as software interrupts if they are generated due to the execution of a software instruction internal to the microcontroller. These are generated when the xxIF flag is set through a software instruction like BSF T1CON, TMR1IF which sets the timer 1 overflow flag when the required conditions for generation of interrupt are met. All the flags related to various modules may be set using such instructions and hence all interrupts in PIC18 devices may be considered software interrupts.

Internal Interrupts

Interrupts are termed as internal interrupts if they are generated due to a module internal to the microcontroller. All the interrupts except for PORTB change and INTx Interrupts that are generated due to port B pins RB0:RB7 are considered internal interrupts. Hence the rest of the interrupts (except RB0:RB7 related) in PIC18 devices may be considered as internal interrupts.

External Interrupts

Interrupts are termed as external interrupts if they are generated due to a signal module external to the microcontroller. The interrupts PORTB change, INTx Interrupts that are generated due to port B pins RB0:RB7 and Reset are considered external interrupts.

Maskable Interrupts

Interrupts are termed as maskable interrupts if they could be stopped from being generated inside a microcontroller due to separate mask bits or enable bits. All the interrupts of PIC18 devices excluding the RESET are considered as maskable interrupts.

Nonmaskable Interrupts:

Interrupts are termed as nonmaskable interrupts if they could not be stopped from being generated inside a microcontroller due to any control/enable/mask bits. RESET is the only nonmaskable interrupt in PIC18 devices as all the other have respective individual or global control/enable bits.

Interrupt Response Sequence

When an interrupt occurs after meeting all the requirements, a sequence of steps takes place. They are as follows.

- 1) Finish the execution of current instruction
- 2) Global Interrupt Enable bit GIE (if IPEN is cleared) is cleared to disable further interrupts. GIE & PEIE/GIEL bits are cleared I (IPEN is set)
- 3) Save the return address (address of next instruction that should have been executed if interrupt hadn't occurred) on to stack. Additionally, WREG, Status & BSR are stored on the fast return stack.
- 4) Copy the address of interrupt vector location (0x08 for high priority and 0x18 for low priority) to PC so that execution is transferred from main program to the corresponding ISR.
- 5) Once in the ISR – interrupt service routine, context save may be performed for the required registers (if required). i.e. saving of the other SFRs or File Registers may also be required depending on the user's application.
- 6) The corresponding interrupt flag bits must be cleared by user's software before re-enabling interrupts. This must be performed before returning from ISR to avoid the recursive interrupts and ensure proper operation.
- 7) The instruction RETFIE – "Return from fast Interrupt Enable" may be used to return from ISR back to main program. This sets the GIE bit (GIE & GIEL/PEIE bit if IPEN is set), in turn re-enable the interrupts. This instruction copies the stored return address back to PC so that execution from ISR is transferred back to instructions of main program (address of next instruction that should have been executed if interrupt hadn't occurred, i.e. the last content of PC before entering the ISR).

When a fast return is not used, it is the responsibility of the user to include instruction to save the WREG, Status & BSR before the actual ISR's instructions are executed. Additionally, these must be restored back to the respective registers before returning from the interrupts ISR.

One common feature of the three PIC18 devices is that there can be a maximum of two priority levels – High Priority with Vector location 0x08 and Low Priority with vector location 0x18. All the interrupts vector to one of these two locations.

Unlike some microcontrollers, the execution is automatically not transferred to the specific dedicated interrupts vector location. Instead, any generated interrupt transfers the execution sequence to one the two vector locations. The capability to identify one interrupt from another vectoring to same vector location is not included in PIC18. Hence automatic clearing of respective flags is not present, and the user must clear the respective flag through the software instructions of the corresponding ISR. This requires determination of the actual interrupt that occurred by polling - checking the flag status in the ISR. An example of the same is presented in program 7.7 of PORTB change interrupt.

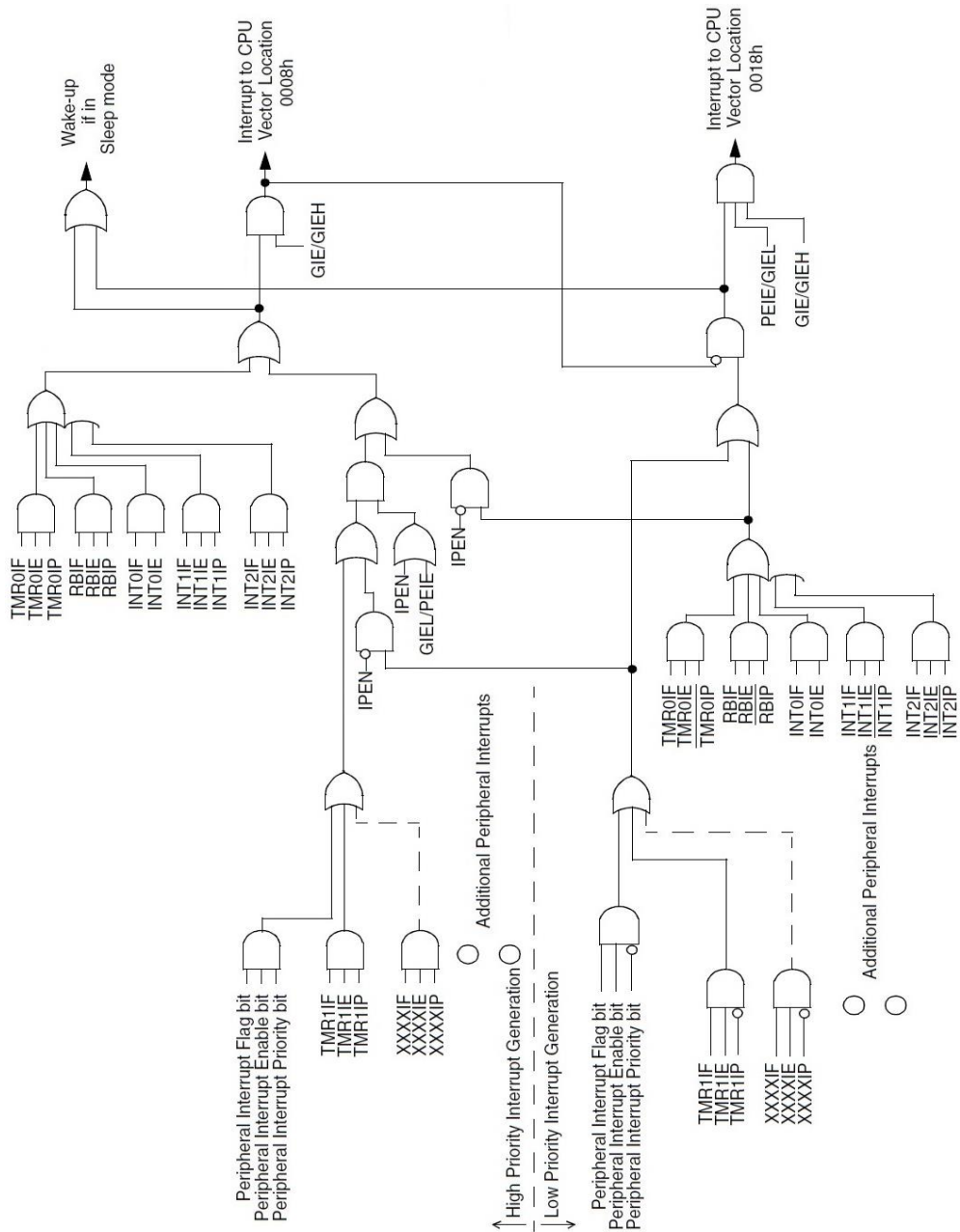


Fig 7.10 Interrupt structure of PIC18F458

Source: Microchip Document: PIC18FXX8 Data Sheet - DS41159E

Table 7.5: Interrupt sources in PIC18F458, PIC18C801 & PIC18F8720

S.No	Module	Name	PIC18F458		PIC18C801		PIC18F8720	
			Source	No	Source	No	Source	No
1	Reset	MCLR, WDT, Reset Instruction, POR, BOR	Reset	1	Reset	1	Reset	1
2	USART1	USART1 Receive Interrupt Flag	RCIF	2	RCIF	2	RC1IF	2
3		USART1 Transmit Interrupt Flag	TXIF	3	TXIF	3	TX1IF	3
4	USART2	USART2 Receive Interrupt Flag	-	-	-	-	RC2IF	4
5		USART2 Transmit Interrupt Flag	-	-	-	-	TX2IF	5
6	Timer 0	TMR0 Overflow Interrupt Flag	TMR0IF	4	TMR0IF	4	TMR0IF	6
7	Timer 1	TMR1 Overflow Interrupt Flag	TMR1IF	5	TMR1IF	5	TMR1IF	7
8	Timer 2	TMR2 Overflow Interrupt Flag	TMR2IF	6	TMR2IF	6	TMR2IF	8
9	Timer 3	TMR3 Overflow Interrupt Flag	TMR3IF	7	TMR3IF	7	TMR3IF	9
10	Timer 4	TMR4 Overflow Interrupt Flag	-	-	-	-	TMR4IF	10
11	ECCP	ECCP1 Interrupt Flag	ECCP1IF	8	-	-	-	-
12	CCP1	CCP1 Interrupt Flag	CCP1IF	9	CCP1IF	8	CCP1IF	11
13	CCP2	CCP2 Interrupt Flag	-	-	CCP2IF	9	CCP2IF	12
14	CCP3	CCP3 Interrupt Flag	-	-	-	-	CCP3IF	13
15	CCP4	CCP4 Interrupt Flag	-	-	-	-	CCP4IF	14
16	CCP5	CCP5 Interrupt Flag	-	-	-	-	CCP5IF	15
17	ADC	A/D Converter Interrupt Flag	ADIF	10	ADIF	10	ADIF	16
18	Comparator	Comparator Interrupt Flag	CMIF	11	-	-	CMIF	17
19	EEPROM	EEPROM Write Operation Interrupt Flag	EEIF	12	-	-	EEIF	18
20	PORTB RB4-RB7: (IOC)	RB Port Change Interrupt Flag	RBIF	13	RBIF	11	RBIF	19
21	External INT0	INT0 External Interrupt Flag	INT0IF	14	INT0IF	12	INT0IF	20
22	External INT1	INT1 External Interrupt Flag	INT1IF	15	INT1IF	13	INT1IF	21
23	External INT2	INT2 External Interrupt Flag	INT2IF	16	INT2IF	14	INT2IF	22
24	External INT3	INT3 External Interrupt Flag	-	-	-	-	INT3IF	23
25	Low Voltage Detect	Low-Voltage Detect Interrupt Flag	LVDIF	17	LVDIF	15	LVDIF	24
26	PSP	Parallel Slave Port Read/Write Interrupt Flag	PSPIF	18	-	-	PSPIF	25
27	MSSP: I2C	Bus Collision Interrupt Flag	BCLIF	19	BCLIF	16	BCLIF	26
28	MSSP	Master Synchronous Serial Port Interrupt Flag	SSPIF	20	SSPIF	17	SSPIF	27
29	CAN	CAN Invalid Received Message Interrupt Flag	IRXIF	21	-	-	-	-
30		CAN bus Activity Wake-up Interrupt Flag	WAKIF	22	-	-	-	-
31		CAN bus Error Interrupt Flag	ERRIF	23	-	-	-	-
32		CAN Transmit Buffer 2 Interrupt Flag	TXB2IF	24	-	-	-	-
33		CAN Transmit Buffer 1 Interrupt Flag	TXB1IF	25	-	-	-	-
34		CAN Transmit Buffer 0 Interrupt Flag	TXB0IF	26	-	-	-	-
35		CAN Receive Buffer 1 Interrupt Flag	RXB1IF	27	-	-	-	-
36		CAN Receive Buffer 0 Interrupt Flag	RXB0IF	28	-	-	-	-
	Additional factors that effect program execution sequence							
	Reset/Power Status	Watchdog Time Out Flag	TO	-	TO	-	TO	-
		Reset Instruction Flag	RI	-	RI	-	RI	-
		Power Down Detection Flag	PD	-	PD	-	PD	-
		Power on Reset Status	POR	-	POR	-	POR	-
		Brown out Reest Status	BOR	-	-	-	BOR	-

The polling used in interrupts is different to the polling required when interrupts are not used. Here in the ISR the respective flag is checked only once and then may skip to checking the next flag and so on based on requirement. Later after the ISR, the PIC18 may be put to sleep whereas if interrupts were not used, the CPU is kept busy checking always and PIC18 could not be put to sleep due to which power consumption is higher. The power may be saved when device is put to sleep and only woken up by an interrupt.

Additionally, the temporary stack allocated for context saving (WREG, Status & BSR) is only one level deep and nesting of interrupts without a context save through user's software may result in an undesired operation.

The return address stack is limited to 31 levels and is used for program call and interrupts. Hence multiple nesting of interrupts and programs calls must ensure to keep within the 31 levels of stack. i.e. total maximum of 31 nested interrupts/program calls.

When priority levels are enabled a high priority interrupt source can interrupt the current execution of an interrupt response of a low priority source. Hence a maximum of 29 levels of calls are allowed when interrupts are used with priority levels. For interrupts with single priority level or priorities disabled 30 levels of call are allowed and when interrupts are disabled a maximum 31 levels of call are allowed.

Latency is another factor that must be considered. Latency varies from one to four instruction cycles based on the type of interrupt. Typically, it is 1-2 instruction cycles for internal interrupts and for external interrupts such as PORTB input change and INTx interrupt the latency is around 3-4 instruction cycles. Also, it is recommended that MOVFF instruction is not used to update the interrupt control registers when any interrupt is enabled as this may cause an undesired response.

An overview of the interrupt structure is shown in fig 7.10. The different interrupt sources and the corresponding interrupt flags for the three devices are shown in Table 7.5.

7.5.1 Registers related to Interrupts

Various registers are provided for control and functioning of interrupts. The Table 7.6 presents the summary of registers that are available in different devices related to interrupts. The control/operational bits of the respective registers for a specific PIC18 is presented here.

Interrupt Flag Bits

Irrespective of the PIC18 device and the register they are associated with, the interrupt flag bit when set to 1 indicates that the corresponding event has occurred. When the respective interrupt flag bit is set to 1 due to an event or software instruction and when individual and global interrupts are enabled, an interrupt would be generated, and execution is transferred to ISR. The respective interrupt flag bit must be cleared to 0 in ISR, before re-

enabling of the respective interrupts (except for RCxIF and TXxIF, where RCREGx is cleared only upon read of RCREGx or and TXIFx is cleared only upon write to TXREGx). The interrupt flag bits of various modules are presented below.

- External Interrupt flag bits: RBIF, INTOIF, INT1IF, INT2IF, INT3IF,
- Timer Interrupt flag bits: TMR0IF, TMR1IF, TMR2IF, TMR3IF, TMR4IF,
- Capture, Compare PWM Interrupt flag bits: CCP1IF, CCP2IF, CCP3IF, CCP4IF, CCP5IF, EEP1IF
- EEPROM & Low Voltage Detect Interrupt flag bits: EEIF, LVDIF,
- ADC, Comparator & Parallel slave port Interrupt flag bits: ADIF, CMIF, PSPIF
- Serial Communication Interrupt flag bits: RCIF, RC1IF, RC2IF, TXIF, TX1IF, TX2IF, BCLIF, SSPIF
- CAN Interrupt flag bits: IRXIF, WAKIF, ERRIF, TXB2IF, TXB1IF, TXB0IF, RXB1IF, RXB0IF

Table 7.6: Interrupt Registers Summary

Register Name	Register	PIC18F458	PIC18C801	PIC18F8720
Interrupt Control Register	INTCON	Yes	Yes	Yes
	INTCON2	Yes	Yes	Yes
	INTCON3	Yes	Yes	Yes
Peripheral Interrupt Request Register	PIR1	Yes	Yes	Yes
	PIR2	Yes	Yes	Yes
	PIR3	Yes	No	Yes
Peripheral Interrupt Enable Register	PIE1	Yes	Yes	Yes
	PIE2	Yes	Yes	Yes
	PIE3	Yes	No	Yes
Peripheral Interrupt Priority Register	IPR1	Yes	Yes	Yes
	IPR2	Yes	Yes	Yes
	IPR3	Yes	No	Yes
Reset Control Register	RCON	Yes	Yes	Yes

Note: PIC18C801 does not have the PIR3, PIE3 and IPR3 registers. They are present in the other two PIC18's. All the interrupt control, peripheral interrupt request, peripheral interrupt enable and peripheral interrupt priority registers are readable and writeable.

Interrupt Enable Bits - Individual

Irrespective of the PIC18 device and the register they are associated with, the interrupt Enable bit when set to 1 enables the corresponding individual interrupt. The respective interrupt may only be generated if the corresponding individual interrupt is enabled through the enable bit by software instruction and when individual and global interrupts are enabled. When an event or a software instruction causes a flag to be set, an interrupt would be generated, and execution is transferred to ISR. The respective interrupt enable bit must be

cleared to 0 to disable the corresponding interrupt. The individual interrupt enable bits of various modules are presented below.

- External Interrupt enable bits: RBIE, INTOIE, INT1IE, INT2IE, INT3IE
- Timer interrupt enable bits: TMR0IE, TMR1IE, TMR2IE, TMR3IE, TMR4IE
- CCP interrupt enable bits: CCP1IE, CCP2IE, CCP3IE, CCP4IE, CCP5IE, EEP1IE
- EEPROM & Low Voltage Detect Interrupt Enable bits: EEIE, LVDIE
- ADC, Comparator & Parallel slave port Interrupt enable bits: ADIE, CMIE, PSPIE
- Serial Communication Interrupt enable bits: RCIE, RC1IE, RC2IE, TXIE, TX1IE, TX2IE, BCLIE, SSPIE
- CAN Interrupt enable bits: IRXIE, WAKIE, ERRIE, TXB2IE, TXB1IE, TXB0IE, RXB1IE, RXB0IE

The details of the Global Interrupt Enable Bit – GIE/GIEH, Priority Levels Enable IPEN, and Peripheral Interrupt Enable Bit – PEIE/GIEL are presented earlier.

Interrupt Priority Bits

The value of the priority bits is irrelevant when IPEN bit in RCON register is cleared to 0. When IPEN bit in RCON register is set, irrespective of the PIC18 device and the register they are associated with, the interrupt Priority bit when set to 1 makes the corresponding individual interrupt a high priority interrupt. The respective interrupt is made a low priority when the priority bit is cleared to 0. When set to 1 and made a high priority interrupt, when interrupt is generated, (if the required conditions are met) execution is transferred to high priority vector location 0x08 and when an interrupt is set to a low priority the execution is transferred to a low priority vector location 0x18. The individual interrupt priority bits of various modules are presented below.

- External Interrupt priority bits: RBIP, INT1IP, INT2IP, INT3IP
- Timer interrupt priority bits: TMR0IP, TMR1IP, TMR2IP, TMR3IP, TMR4IP
- CCP interrupt priority bits: CCP1IP, CCP2IP, CCP3IP, CCP4IP, CCP5IP, EEP1IP
- EEPROM & Low Voltage Detect interrupt priority bits: EEIP, LVDIP
- ADC, Comparator & Parallel slave port Interrupt priority bits: ADIP, CMIP, PSPIP
- Serial Communication Interrupt priority bits: RCIP, RC1IP, RC2IP, TXIP, TX1IP, TX2IP, BCLIP, SSPIP
- CAN Interrupt priority bits: IRXIP, WAKIP, ERRIP, TXB2IP, TXB1IP, TXB0IP, RXB1IP, RXB0IP

Note that Interrupt priority for external interrupt INT0 doesn't exist in all three PIC18 devices as it is always a high priority interrupt.

The control/operational bits of interrupt related registers for specific PIC18 are presented here: PIC18F458 in Table 7.7, PIC18C801 in Table 7.8 & PIC18F8720 in Table 7.9.

Table 7.7: Registers related to Interrupts of PIC18F458

Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
INTCON	GIE/GIEH	PEIE/GIEL	TMR0IE	INT0IE	RBIE	TMR0IF	INT0IF	RBIF
INTCON2	RBP $\bar{U}$	INTEDG0	INTEDG1	INTEDG2	—	TMR0IP	—	RBIP
INTCON3	INT2IP	INT1IP	—	INT2IE	INT1IE	—	INT2IF	INT1IF
PIR1	PSPIF	ADIF	RCIF	TXIF	SSPIF	CCP1IF	TMR2IF	TMR1IF
PIR2	—	CMIF	—	EEIF	BCLIF	LVDIF	TMR3IF	ECCP1IF
PIR3	IRXIF	WAKIF	ERRIF	TXB2IF	TXB1IF	TXB0IF	RXB1IF	RXB0IF
PIE1	PSPIE	ADIE	RCIE	TXIE	SSPIE	CCP1IE	TMR2IE	TMR1IE
PIE2	—	CMIE	—	EEIE	BCLIE	LVDIE	TMR3IE	ECCP1IE
PIE3	IRXIE	WAKIE	ERRIE	TXB2IE	TXB1IE	TXB0IE	RXB1IE	RXB0IE
IPR1	PSPIP	ADIP	RCIP	TXIP	SSPIP	CCP1IP	TMR2IP	TMR1IP
IPR2	—	CMIP	—	EEIP	BCLIP	LVDIP	TMR3IP	ECCP1IP
IPR3	IRXIP	WAKIP	ERRIP	TXB2IP	TXB1IP	TXB0IP	RXB1IP	RXB0IP
RCON	IPEN	—	—	R $\bar{I}$	T $\bar{O}$	PD	POR	BOR

Table 7.8: Registers related to Interrupts of PIC18C801

Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
INTCON	GIE/GIEH	PEIE/GIEL	TMR0IE	INT0IE	RBIE	TMR0IF	INT0IF	RBIF
INTCON2	RBP $\bar{U}$	INTEDG0	INTEDG1	INTEDG2	—	TMR0IP	—	RBIP
INTCON3	INT2IP	INT1IP	—	INT2IE	INT1IE	—	INT2IF	INT1IF
PIR1	—	ADIF	RCIF	TXIF	SSPIF	CCP1IF	TMR2IF	TMR1IF
PIR2	—	—	—	—	BCLIF	LVDIF	TMR3IF	CCP2IF
PIE1	—	ADIE	RCIE	TXIE	SSPIE	CCP1IE	TMR2IE	TMR1IE
PIE2	—	—	—	—	BCLIE	LVDIE	TMR3IE	CCP2IE
IPR1	—	ADIP	RCIP	TXIP	SSPIP	CCP1IP	TMR2IP	TMR1IP
IPR2	—	—	—	—	BCLIP	LVDIP	TMR3IP	CCP2IP
RCON	IPEN	r	—	R $\bar{I}$	T $\bar{O}$	PD	POR	r

Table 7.9: Registers related to Interrupts of PIC18F8720

Register	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
INTCON	GIE/GIEH	PEIE/GIEL	TMR0IE	INT0IE	RBIE	TMR0IF	INT0IF	RBIF
INTCON2	RBP $\bar{U}$	INTEDG0	INTEDG1	INTEDG2	INTEDG3	TMR0IP	INT3IP	RBIP
INTCON3	INT2IP	INT1IP	INT3IE	INT2IE	INT1IE	INT3IF	INT2IF	INT1IF
PIR1	PSPIF	ADIF	RC1IF	TX1IF	SSPIF	CCP1IF	TMR2IF	TMR1IF
PIR2	—	CMIF	—	EEIF	BCLIF	LVDIF	TMR3IF	CCP2IF
PIR3	—	—	RC2IF	TX2IF	TMR4IF	CCP5IF	CCP4IF	CCP3IF
PIE1	PSPIE	ADIE	RC1IE	TX1IE	SSPIE	CCP1IE	TMR2IE	TMR1IE
PIE2	—	CMIE	—	EEIE	BCLIE	LVDIE	TMR3IE	CCP2IE
PIE3	—	—	RC2IE	TX2IE	TMR4IE	CCP5IE	CCP4IE	CCP3IE
IPR1	PSPIP	ADIP	RC1IP	TX1IP	SSPIP	CCP1IP	TMR2IP	TMR1IP
IPR2	—	CMIP	—	EEIP	BCLIP	LVDIP	TMR3IP	CCP2IP
IPR3	—	—	RC2IP	TX2IP	TMR4IP	CCP5IP	CCP4IP	CCP3IP
RCON	IPEN	—	—	R $\bar{I}$	T $\bar{O}$	PD	POR	BOR

The highlighted bits indicate that there is a variation in the respective bits of different devices – the PIC18F458, PIC18C801 and PIC18F8720.

Other operational control bits associated with interrupt registers

- RBPU, INTEDG0, INTEDG1, INTEDG2, INTEDG3 (refer External Interrupts: Ch 7.6)
- RI, TO, PD, POR, BOR (refer Special Features Ch 17)

RBPU - PORTB Pull-up Enable bit

All pins of PORTB are equipped with a weak internal pull-up. The RBPU acts as a single control pin to enable pull-ups for all the pins of PORTB.

When the RBPU bit in INTCON2 register when set to 0 disables the pull-ups and when set to 1 enables pull-ups based on the value of corresponding latch value. When the port pins are configured as outputs or upon a POR, the weak pull-up is automatically disabled. Upon a POR the pull up is disabled by default as RBPU is set to 1.

INTEDG0:INTEDG3 – External Interrupt 0: External Interrupt 3 Edge Select bit

The PORTB also consists of external interrupts INT0:INT3. Note that INT3, INTEDG3 is only present in PIC18F8720. The external interrupts could be triggered on a rising or falling edge at RB0:RB3 pins respectively. The selection for interrupt generation of falling or rising edge on RB0:RB3 is configured through the INTEDG0:INTEDG3 bits of INTCON2 register. When the INTEDGx bit (x represents number between 0 to 3) is set to 1 interrupt is generated upon a rising edge and when set to 0 interrupt is generated on a falling edge.

RI, TO, PD, POR, BOR: Power and Reset related status bits

The Power and Reset related status bits are associated with RCON - Reset Control register. A brief overview of the bits is presented here, and detailed information is presented in chapter 16.

The RI indicates the RESET Instruction Flag bit. This is a status bit and indicates whether RESET instruction is executed or not. It remains set to 1 when RESET instruction is not executed & is set to 0 when a RESET instruction is executed, causing a device reset.

The TO indicate the Watchdog Time-out Flag bit. This is a status bit and indicates when whether watch dog time is completed or not. It is set to 1 and stays set to 1 upon power-up or execution of CLRWDT or sleep instruction. It is set to 0 when the assigned time of watchdog timer is completed (time-out).

The PD indicates the Power-Down Detection Flag bit. This is a status bit and indicates whether the PIC18 is in power down mode. It is set to 1 and remains set upon power-up or by the execution of CLRWDT instruction. It is cleared to 0 upon the execution of SLEEP instruction.

The POR indicates the Power-on Reset status bit. This is a status bit and indicates whether a Power-on Reset has occurred or not. It is set to 1 and remains set if Power-on Reset has not occurred and clears to 0 when a Power-on Reset occurred. This bit must be set after a POR occurs.

Table 7.10: Interrupt Flag, Enable & Priority bits and respective registers of PIC18F458

S.No	Interrupt Name	PIC18F458						
		No	Flag	Flag Reg	Enable	Enable Reg	Priority	Priority Reg
1	Reset	1	Reset	-	-	-	-	-
2	USART1 Receive	2	RCIF	PIR1	RCIE	PIE1	RCIP	IPR1
3	USART1 Transmit	3	TXIF	PIR1	TXIE	PIE1	TXIP	IPR1
4	USART2 Receive	-	-	-	-	-	-	-
5	USART2 Transmit	-	-	-	-	-	-	-
6	TMR0 Overflow	4	TMR0IF	INTCON	TMR0IE	INTCON	TMR0IP	INTCON2
7	TMR1 Overflow	5	TMR1IF	PIR1	TMR1IE	PIE1	TMR1IP	IPR1
8	TMR2 Overflow	6	TMR2IF	PIR1	TMR2IE	PIE1	TMR2IP	IPR1
9	TMR3 Overflow	7	TMR3IF	PIR2	TMR3IE	PIE2	TMR3IP	IPR2
10	TMR4 Overflow	-	-	-	-	-	-	-
11	ECCP1	8	ECP1IF	PIR2	ECCP1IE	PIE2	ECCP1IP	IPR2
12	CCP1	9	CCP1IF	PIR1	CCP1IE	PIE1	CCP1IP	IPR1
13	CCP2	-	-	-	-	-	-	-
14	CCP3	-	-	-	-	-	-	-
15	CCP4	-	-	-	-	-	-	-
16	CCP5	-	-	-	-	-	-	-
17	A/D Converter	10	ADIF	PIR1	ADIE	PIE1	ADIP	IPR1
18	Comparator	11	CMIF	PIR2	CMIE	PIE2	CMIP	IPR2
19	EEPROM Write Operation	12	EEIF	PIR2	EEIE	PIE2	EEIP	IPR2
20	RB Port Change	13	RBIF	INTCON	RBIE	INTCON	RBIP	INTCON2
21	INT0 External	14	INT0IF	INTCON	INT0IE	INTCON	-	-
22	INT1 External	15	INT1IF	INTCON3	INT1IE	INTCON3	INT1IP	INTCON3
23	INT2 External	16	INT2IF	INTCON3	INT2IE	INTCON3	INT2IP	INTCON3
24	INT3 External	-	-	-	-	-	-	-
25	Low-Voltage Detect	17	LVDIF	PIR2	LVDIE	PIE2	LVDIP	IPR2
26	Parallel Slave Port R/W	18	PSPIF	PIR1	PSPIE	PIE1	PSPIP	IPR1
27	Bus Collision	19	BCLIF	PIR2	BCLIE	PIE2	BCLIP	IPR2
28	Master Synchronous Serial Port	20	SSPIF	PIR1	SSPIE	PIE1	SSPIP	IPR1
29	CAN Invalid Received Message	21	IRXIF	PIR3	IRXIE	PIE3	IRXIP	IPR3
30	CAN bus Activity Wake-up	22	WAKIF	PIR3	WAKIE	PIE3	WAKIP	IPR3
31	CAN bus Error	23	ERRIF	PIR3	ERRIE	PIE3	ERRIP	IPR3
32	CAN Transmit Buffer 2	24	TXB2IF	PIR3	TXB2IE	PIE3	TXB2IP	IPR3
33	CAN Transmit Buffer 1	25	TXB1IF	PIR3	TXB1IE	PIE3	TXB1IP	IPR3
34	CAN Transmit Buffer 0	26	TXB0IF	PIR3	TXB0IE	PIE3	TXB0IP	IPR3
35	CAN Receive Buffer 1	27	RXB1IF	PIR3	RXB1IE	PIE3	RXB1IP	IPR3
36	CAN Receive Buffer 0	28	RXB0IF	PIR3	RXB0IE	PIE3	RXB0IP	IPR3

The BOR indicates the Brown-out Reset Status bit. This is not present in PIC18C801. This is a status bit and indicates whether a Brown-out Reset has occurred or not. It is set to 1 and remains set to 1 when a Brown-out has not occurred and is set to 0 when brown-out occurred. This bit must be set after a BOR occurs.

The three required bits (Flag, Enable & Priority) and related register for each associated interrupt source is presented here: PIC18F458 in Table 7.10, PIC18C801 in Table 7.11 & PIC18F8720 in Table 7.12.

Table 7.11: Interrupt Flag, Enable & Priority bits and respective registers of PIC18C801

S.No	Interrupt Name	PIC18C801						
		No	Flag	Flag Reg	Enable	Enable Reg	Priority	Priority Reg
1	Reset	1	Reset	-	-	-	-	-
2	USART1 Receive	2	RCIF	PIR1	RCIE	PIE1	RCIP	IPR1
3	USART1 Transmit	3	TXIF	PIR1	TXIE	PIE1	TXIP	IPR1
4	USART2 Receive	-	-	-	-	-	-	-
5	USART2 Transmit	-	-	-	-	-	-	-
6	TMR0 Overflow	4	TMR0IF	INTCON	TMR0IE	INTCON	TMR0IP	INTCON2
7	TMR1 Overflow	5	TMR1IF	PIR1	TMR1IE	PIE1	TMR1IP	IPR1
8	TMR2 Overflow	6	TMR2IF	PIR1	TMR2IE	PIE1	TMR2IP	IPR1
9	TMR3 Overflow	7	TMR3IF	PIR2	TMR3IE	PIE2	TMR3IP	IPR2
10	TMR4 Overflow	-	-	-	-	-	-	-
11	ECCP1	-	-	-	-	-	-	-
12	CCP1	8	CCP1IF	PIR1	CCP1IE	PIE1	CCP1IP	IPR1
13	CCP2	9	CCP2IF	PIR2	CCP2IE	PIE2	CCP2IP	IPR2
14	CCP3	-	-	-	-	-	-	-
15	CCP4	-	-	-	-	-	-	-
16	CCP5	-	-	-	-	-	-	-
17	A/D Converter	10	ADIF	PIR1	ADIE	PIE1	ADIP	IPR1
18	Comparator	-	-	-	-	-	-	-
19	EEPROM Write Operation	-	-	-	-	-	-	-
20	RB Port Change	11	RBIF	INTCON	RBIE	INTCON	RBIP	INTCON2
21	INT0 External	12	INT0IF	INTCON	INT0IE	INTCON	-	-
22	INT1 External	13	INT1IF	INTCON3	INT1IE	INTCON3	INT1IP	INTCON3
23	INT2 External	14	INT2IF	INTCON3	INT2IE	INTCON3	INT2IP	INTCON3
24	INT3 External	-	-	-	-	-	-	-
25	Low-Voltage Detect	15	LVDIF	PIR2	LVDIE	PIE2	LVDIP	IPR2
26	Parallel Slave Port R/W	-	-	-	-	-	-	-
27	Bus Collision	16	BCLIF	PIR2	BCLIE	PIE2	BCLIP	IPR2
28	Master Synchronous Serial Port	17	SSPIF	PIR1	SSPIE	PIE1	SSPIP	IPR1
29	CAN Invalid Received Message	-	-	-	-	-	-	-
30	CAN bus Activity Wake-up	-	-	-	-	-	-	-
31	CAN bus Error	-	-	-	-	-	-	-
32	CAN Transmit Buffer 2	-	-	-	-	-	-	-
33	CAN Transmit Buffer 1	-	-	-	-	-	-	-
34	CAN Transmit Buffer 0	-	-	-	-	-	-	-
35	CAN Receive Buffer 1	-	-	-	-	-	-	-
36	CAN Receive Buffer 0	-	-	-	-	-	-	-

Table 7.12: Interrupt Flag, Enable & Priority bits and respective registers of PIC18F8720

S.No	Interrupt Name	PIC18F8720						
		No	Flag	Flag Reg	Enable	Enable Reg	Priority	Priority Reg
1	Reset	1	Reset	-	-	-	-	-
2	USART1 Receive	2	RC1IF	PIR1	RC1IE	PIE1	RC1IP	IPR1
3	USART1 Transmit	3	TX1IF	PIR1	TX1IE	PIE1	TX1IP	IPR1
4	USART2 Receive	4	RC2IF	PIR3	RC2IE	PIE3	RC2IP	IPR3
5	USART2 Transmit	5	TX2IF	PIR3	TX2IE	PIE3	TX2IP	IPR3
6	TMR0 Overflow	6	TMR0IF	INTCON	TMR0IE	INTCON	TMR0IP	INTCON2
7	TMR1 Overflow	7	TMR1IF	PIR1	TMR1IE	PIE1	TMR1IP	IPR1
8	TMR2 Overflow	8	TMR2IF	PIR1	TMR2IE	PIE1	TMR2IP	IPR1
9	TMR3 Overflow	9	TMR3IF	PIR2	TMR3IE	PIE2	TMR3IP	IPR2
10	TMR4 Overflow	10	TMR4IF	PIR3	TMR4IE	PIE3	TMR4IP	IPR3
11	ECCP1	-	-	-	-	-	-	-
12	CCP1	11	CCP1IF	PIR1	CCP1IE	PIE1	CCP1IP	IPR1
13	CCP2	12	CCP2IF	PIR2	CCP2IE	PIE2	CCP2IP	IPR2
14	CCP3	13	CCP3IF	PIR3	CCP3IE	PIE3	CCP3IP	IPR3
15	CCP4	14	CCP4IF	PIR3	CCP4IE	PIE3	CCP4IP	IPR3
16	CCP5	15	CCP5IF	PIR3	CCP5IE	PIE3	CCP5IP	IPR3
17	A/D Converter	16	ADIF	PIR1	ADIE	PIE1	ADIP	IPR1
18	Comparator	17	CMIF	PIR2	CMIE	PIE2	CMIP	IPR2
19	EEPROM Write Operation	18	EEIF	PIR2	EEIE	PIE2	EEIP	IPR2
20	RB Port Change	19	RBIF	INTCON	RBIE	INTCON	RBIP	INTCON2
21	INT0 External	20	INT0IF	INTCON	INT0IE	INTCON	-	-
22	INT1 External	21	INT1IF	INTCON3	INT1IE	INTCON3	INT1IP	INTCON3
23	INT2 External	22	INT2IF	INTCON3	INT2IE	INTCON3	INT2IP	INTCON3
24	INT3 External	23	INT3IF	INTCON3	INT3IE	INTCON3	INT3IP	INTCON2
25	Low-Voltage Detect	24	LVDIF	PIR2	LVDIE	PIE2	LVDIP	IPR2
26	Parallel Slave Port R/W	25	PSPIF	PIR1	PSPIE	PIE1	PSPIP	IPR1
27	Bus Collision	26	BCLIF	PIR2	BCLIE	PIE2	BCLIP	IPR2
28	Master Synchronous Serial Port	27	SSPIF	PIR1	SSPIE	PIE1	SSPIP	IPR1
29	CAN Invalid Received Message	-	-	-	-	-	-	-
30	CAN bus Activity Wake-up	-	-	-	-	-	-	-
31	CAN bus Error	-	-	-	-	-	-	-
32	CAN Transmit Buffer 2	-	-	-	-	-	-	-
33	CAN Transmit Buffer 1	-	-	-	-	-	-	-
34	CAN Transmit Buffer 0	-	-	-	-	-	-	-
35	CAN Receive Buffer 1	-	-	-	-	-	-	-
36	CAN Receive Buffer 0	-	-	-	-	-	-	-

7.6 External Interrupts & its programming

The interrupts associated with PORTB are external in nature. These are two types.

1) PORTB change or Interrupt on change on RB4-RB7, 2) External interrupts INT0, INT1, INT2 (PIC18F8720 additionally has INT3). These are discussed in the following sections.

7.6.1 Interrupt on Change/PORTB Change

One of the special functionalities of PORTB is the Interrupt on change or the PORTB change feature and it is available for four of the PORTB pins: Pin 4 to Pin 7, i.e RB4-RB7. A typical application of the interrupt on change feature is to wake-up on key depression operation, i.e. keypad interfacing.

All the pins of port B are provided with a weak internal pull-up. An internal pull-up is helpful and is required when the device connected externally to the PORTB input pins is not capable to provide the sufficient voltage to maintain the required levels. The Pull-up is not required if the required levels could be maintained. The pull-up for all the pins of port B is enabled or disabled by a single control bit RBPU. The pull-up is enabled when the RBPU bit in INTCON2 register is cleared to 0 and is disabled when the control bit RBPU is set to 1. When the port pins are configured as outputs or upon a POR, the weak pull-up is automatically disabled. Upon a POR the pull up is disabled by default as RBPU is set to 1.

The interrupt on change feature of PORTB requires that the corresponding port pins are configured as inputs by setting the corresponding TRISB bits to 1 to be able to generate the interrupt upon PORTB change. The TRISB register controls the direction for the corresponding interrupt on change feature.

The pins that are configured as inputs only can generate an interrupt for a PORTB change. A single interrupt is provided for all the four pins. A change in any one of the four input configured pins, RB4 to RB7 generates the same interrupt and sets the RB Port Change Interrupt Flag bit – RBIF in INTCON register. The interrupt on change feature does not work on the pins that are configured as outputs as they will be excluded from interrupt on change comparison. The interrupt structure of PORTB change is shown in fig 7.11. The PORTB change interrupt's related bits and registers for different PIC18 devices are presented in Table 7.13.

The interrupt on change comparison works as follows. The RB4:RB7 input pins are compared with the old value that is latched during a previous read of PORTB. Whenever there is difference between the previous latched value and the current port pin value a change is detected and is considered as a mismatch. An interrupt is generated when there is at least one mismatch. This is generated through an OR operation of all the comparison outputs. This interrupt of change feature allows the microcontroller to wake-up the device CPU from sleep.

Reading the PORTB ends a mismatch (interrupt on change) and clears the condition that sets the RBIF flag. Also, a write operation involves a read. As an example, the "CLR PORTB" instruction reads the PORTB, clears all the bits and then writes the result to PORTB. This reading of PORTB clears the mismatch, also clears the condition that sets the RBIF flag. Hence the LATB must be read instead of PORTB.



PIC18 Device	Interrupt Name	Flag	Flag Reg	Enable	Enable Reg	Priority	Priority Reg	Pull Up	Pull Up Reg
PIC18F458, PIC18C801 & PIC18F8720	RB Port Change	RBIF	INTCON	RBIE	INTCON	RBIP	INTCON2	$\overline{\text{RBPU}}$	INTCON2

The mismatch may be cleared in the following manner by the user in the ISR:

- Any read or write of PORTB except for the MOVFF instruction to end the mismatch condition.
- Clear flag bit RBIF.

Program 7.6: A button/switch is connected to the pin 5 of PORTB and an LED to pin 0 of PORTC. Use the Interrupt on change feature to obtain the status of the button/switch and blink the LED if the switch is pressed.

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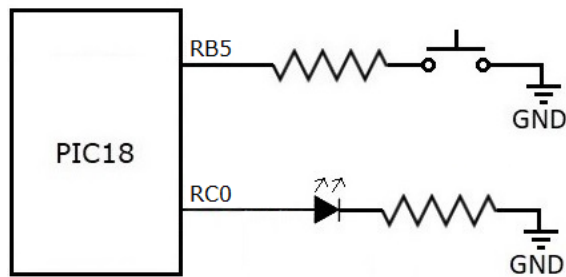


Fig P7.6 Button/Switch & its LED Interfacing

	#INCLUDE <P18F458.INC>; CAN BE OTHER PROCESSOR		
	ORG	0x00	; Set Origin 0x0000 of Flash memory
	GOTO	START	; Go to Start of Main program.
	ORG	0x08	; Set Origin 0x08 for High Priority Interrupt ; and 0x18 for Low Priority Interrupt
	BTFSS	INTCON,RBIF	; Check if interrupt is due to RBIF, ; if its due to RBIF go to RB_ISR
	RETFIE		; If RBIF not set, return from interrupt.
	GOTO	RB_ISR	; Go to RB ISR
	ORG	0x100	; Set Origin 0x100 for RB ISR
RB_ISR	MOVF	PORTB,0	; Read PORTB to WREG, WREG = PORTB
	BCF	INTCON,RBIF	; Clear RBIF flag
	ANDLW	0x20	; Extract status of Port B pin 5
	BZ	IL1	; If RB5 is 0, go to IL1
	BCF	PORTC,0	; RB5 is 1, Button Not Pressed ; So turn OFF the LED at RC0
	BRA	IL2	; Go to IL2
IL1	BSF	PORTC,0	; RB5 is 0, Button Pressed ; So turn ON the LED at RC0
IL2	RETFIE		; Return from fast Interrupt Enable
	ORG	0x200	; Set Origin 0x200 for Main program
START	BSF	TRISB,5	; TRISB.5 = 1, Make PORTB Pin 5 as Input
	BCF	TRISC,0	; TRISC.0 = 0, Make PORTC pin 0 as output
	BCF	INTCON2,RBPU	; Enable the weak pull up for PORTB
	BSF	RCON,IPEN	; Enable Priority levels
	BSF	INTCON2,RBIP	; Set RBIF as high priority
	BCF	INTCON,RBIF	; Clear RBIF flag
	BSF	GIE	; Enable Global Interrupts
	BSF	INTCON,RBIE	; Enable PORTB change interrupt
M1	NOP		; Alternate Logic
	BRA	M1	; Branch to M1 to repeat change state/blinking
	END		

Program 7.7: Four buttons/switches are connected to the pin 4, 5, 6 & 7 of PORTB and two LEDs to pin 0 and 1 of PORTC. The button at pin 4 is denoted on the LEDs as 00, 5 as 01, 6 as 10 and 7 as 11, where 1 represents an ON LED and 0 represents an OFF LED. Use the Interrupt on change feature to obtain the status of the buttons/switches and indicate the button number by turning on the correct combination of LEDs.

This program and setup is similar to that of button example program 7.5. This program does not use weak pull-up provided by PORTB but uses an external pull-down with active high logic for buttons & LEDs. The setup is shown in fig P7.7. Here it is assumed that no two switches are pressed at the same time.

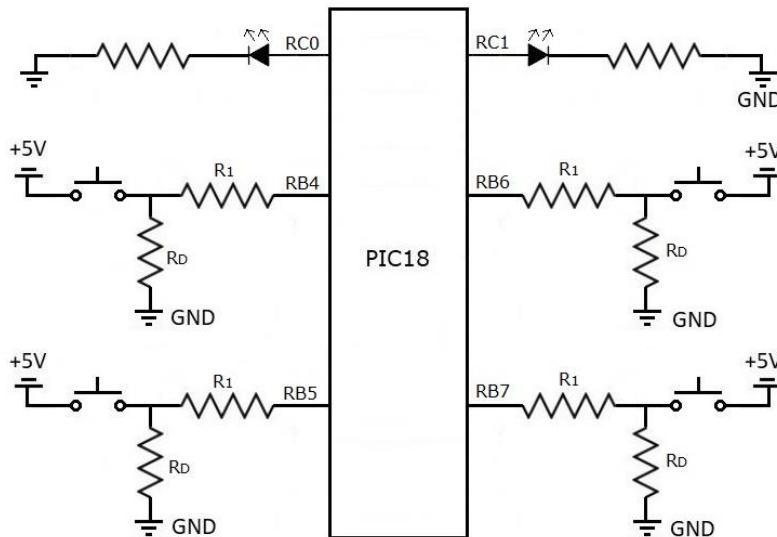


Fig P7.7 Button/Switch & its LED Interfacing

	#INCLUDE <P18F458.INC>; CAN BE OTHER PROCESSOR		
	ORG	0x00	; Set Origin 0x0000 of Flash memory
	GOTO	START	; Go to Start of Main program.
	ORG	0x08	; Set Origin 0x08 for High Priority Interrupt ; and 0x18 for Low Priority Interrupt
	BTFSS	INTCON,RBIF	; Check if interrupt is due to RBIF, if due to RBIF ; go to RB_ISR
	RETFIE		; If RBIF not set, return from interrupt
	GOTO	RB_ISR	; Go to RB ISR
	ORG	0x100	; Set Origin 0x100 for RB ISR
RB_ISR	MOVF	PORTB,0	; Read PORTB to WREG, WREG = PORTB
	BCF	INTCON,RBIF	; Clear RBIF flag
	ANDLW	0xF0,0	; Extract status of Port B change bits, RB4-RB7
	GOTO	BC_LS	; Go to BC_LS
	ORG	0x150	; Set Origin 0x150 for BC_LS module

BC_LS	BTFSC	WREG,4	; Check if interrupt is due to change on Pin 4
	GOTO	L1	; Go to L1
	BTFSC	WREG,5	; Check if interrupt is due to change on Pin 5
	GOTO	L2	; Go to L2
	BTFSC	WREG,6	; Check if interrupt is due to change on Pin 6
	GOTO	L3	; Go to L3
	BTFSC	WREG,7	; Check if interrupt is due to change on Pin 7
	GOTO	L4	; Go to L4
	RETFIE		; Return from fast Interrupt Enable
L1	BCF	PORTC,0	; Set combination for Button at Pin 4, LED 0 OFF
	BCF	PORTC,1	; Set combination for Button at Pin 4, LED 1 OFF
	RETFIE		; Return from fast Interrupt Enable
L2	BSF	PORTC,0	; Set combination for Button at Pin 5, LED 0 ON
	BCF	PORTC,1	; Set combination for Button at Pin 5, LED 1 OFF
	RETFIE		; Return from fast Interrupt Enable
L3	BCF	PORTC,0	; Set combination for Button at Pin 6, LED 0 OFF
	BSF	PORTC,1	; Set combination for Button at Pin 6, LED 1 ON
	RETFIE		; Return from fast Interrupt Enable
L4	BSF	PORTC,0	; Set combination for Button at Pin 7, LED 0 ON
	BSF	PORTC,1	; Set combination for Button at Pin 7, LED 1 ON
	RETFIE		; Return from fast Interrupt Enable
	ORG	0x200	; Set Origin 0x200 for Main program
START	SETF	TRISB,0	; Set all the pins of PORT B as I/Ps, TRISB = 0xFF
	CLRF	TRISC,0	; Set all the pins of PORT C as O/Ps, TRISC = 0x00
	BSF	RCON,IPEN	; Enable Priority levels
	BSF	INTCON2,RBIP	; Set RBIF as high priority
	BCF	INTCON,RBIF	; Clear RBIF flag
	BSF	GIE	; Enable Global Interrupts
	BSF	INTCON,RBIE	; Enable PORTB change interrupt
M1	NOP		; No Operation
	BRA	M1	; Branch to M1 to repeat change state/blinking
	END		

7.6.2 External Interrupts

One of the functionalities of PORTB is the Interrupt on change or the PORTB change feature. The other Interrupt related functionality is the external interrupt feature available at PORTB pins: Pin 0 to Pin 3, i.e. RB0-RB3 as INT0, INT1, INT2 (and INT3 only on PIC18F8720). The external interrupt structure is shown in fig 7.12. The external interrupt related bits and registers for different PIC18 devices are presented in Table 7.14.

Unlike the PORTB change feature that has one interrupt for all the 4 pins of PORTB: RB4-RB7, each of the external interrupts INTx are provided a separate interrupt. Here x can

take any value from 0 to 3, with INT3 being available only on PIC18F8720. Similarly, these external interrupts also have the flexibility to be triggered based on positive or negative edge which is not available with PORTB change. The external interrupts INTx are preferred when more flexibility is required with separate interrupts.

The INTEDGx bit in INTCON2 register provides the software programming flexibility to users to configure and make the INTx external interrupts to be triggered based on negative/falling (high to low) edge or a positive/rising (low to high) edge. When the INTEDGx bit is set to 1 rising edge is selected and clearing to 0 selects a falling edge.

An interrupt flag INTxIF is generated when the corresponding edge is detected at the respective RBx/INTx pin. The corresponding INTxIF interrupts are enabled or disabled through the INTxIE bit. Setting this bit to 1 enables it and clearing to 0 disables the interrupt. Similar to any other interrupt flag in PIC18, the INTxIF bit must be cleared by user's software (i.e in ISR if interrupts are enabled or in main program if it is based on polling) before re-enabling the interrupt.

The INTxIF interrupts priority is set through INTxIP bit and setting this bit to 1 set it as a high priority and clearing to 0 sets it as a low priority interrupt. Apart from INTOIF which is always set to high priority (has no configurable priority bit), all the other interrupts INT1IF, INT2IF & INT3IF could be set to any priority level using the respective INT1IP, INT2IP & INT3IP bits. The INTxIF, INTxIE & INTxIP are accessible through the INTCONy register (i.e INTCON, INTCON2 or INTCON3).

Similar to all the other interrupts, for the external interrupts INTxIF's to work the GIE - Global Interrupt Enable bit must be set for a high priority interrupt and GIEL/PEIE must also be set if the INTxIF interrupts are configured as low priority interrupts. If the required enable bits are set, the processors will branch to the respective priority vector location (0x08 for high priority or 0x18 for low priority).

Similar to RBIF, the external interrupts INTx interrupts can wake-up the CPU of the microcontroller while in sleep mode and branch to its ISR if the respective interrupts were enabled (INTxIE bit set) before the CPU is sent to Sleep. Upon a POR the INTx external interrupts are configured to be triggered based on (low to high) rising edge. Edge detection requires minimum of two instruction cycles each at both the high/logic 1 and low/logic 0 levels. Before a positive edge occurs, the pin level should be held low for a minimum period of 2 instruction cycles and after the occurrence of positive edge the pin level should be held high for minimum period of 2 instruction cycles. i.e. a specific level should be maintained for a minimum of two instruction cycles before and after the occurrence of an edge.

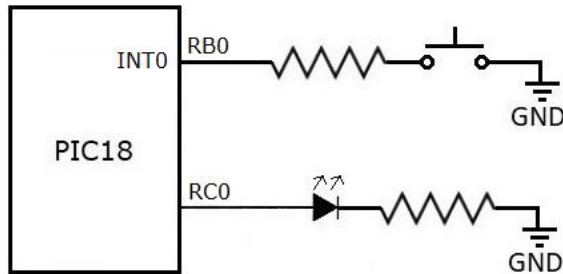


Fig P7.8 Button/Switch & its LED Interfacing

This program and setup are slightly similar to that of button example program 7.4. This program makes use of weak pull-up provided by PORTB. Hence external pull-up is not required and not used. It uses an active low logic for button and active high logic for LED. Since INT0 is fixed to a high priority and doesn't have a priority control bit, no corresponding bit is used to set its priority. The toggling of Led for 5 presses mimics the activation of the special key functions when the shift key is pressed 5 times on a PC keyboard. The setup is shown in fig P7.8.

	#INCLUDE <P18F458.INC>; CAN BE OTHER PROCESSOR		
	ORG	0x00	; Set Origin 0x0000 of Flash memory
	GOTO	START	; Go to Start of Main program.
	ORG	0x08	; Set Origin 0x08 for High Priority Interrupt ; and 0x18 for Low Priority Interrupt
	BTFSS	INTCON,INT0IF	; Check if interrupt is due to INT0IF, ; if its due to INT0IF go to X0_ISR
	RETFIE		; If INT0IF not set, return from interrupt.
	GOTO	X0_ISR	; Go to X0_ISR
	ORG	0x100	; Set Origin 0x100 for RB ISR
X0_ISR	INCF	0x00,1,0	; Increment file reg 0x00.
	BCF	INTCON,INT0IF	; Clear INT0IF flag
	CPFSEQ	0x05,0	; Check if button is pressed 5 times
	GOTO	IL1	; If button not pressed 5 times, go to IL1
	BTG	PORTC,0	; Toggle LED at RC0 for 5 button press
	CLRF	0x00,0	; Clear File Reg 0x00, [0x00] = 0x00
IL1	RETFIE		; Return from fast Interrupt Enable
	ORG	0x200	; Set Origin 0x200 for Main program
START	BSF	TRISB,0	; TRISB.0 = 1, Make PORTB Pin 0 as Input
	BCF	TRISC,0	; TRISC.0 = 0, Make PORTC pin 0 as output
	CLRF	0x00,0	; Clear File Reg 0x00, [0x00] = 0x00
	BCF	INTCON2,RBPU	; Enable the weak pull up for PORTB
	BCF	INTCON,INT0IF	; Clear INT0IF flag
	BCF	INTCON2, INTEDG0	; Set Falling (High to Low) edge detection
	BSF	GIE	; Enable Global Interrupts

	BSF	INTCON,INT0IE	; Enable Ext Int0
	MOVLW	0x05	; WREG = 0x05
M1	NOP		; No Operation, For Alternate Logic
	BRA	M1	; Branch to M1 to repeat, Wait on INT0
	END		

Program 7.9: Four buttons/switches are connected to the pin 0, 1, 2 & 3 of PORTB and two LEDs to pin 0 and 1 of PORTC. The button at pin 0 is denoted on the LEDs as 00, pin 1 as 01, pin 2 as 10 and pin 3 as 11, where 1 represents an ON LED and 0 represents an OFF LED. Use the external interrupts with positive edge detection to obtain the status of the buttons/switches and indicate the button number by turning on the correct combination of LEDs.

This program and setup is similar to that of button example program 7.5. This program does not use weak pull-up provided by PORTB but uses an external pull-down with active high logic for buttons & LEDs. It uses multiple (all 4 external) interrupts. The setup is shown in fig P7.9. Here it is assumed that no two switches are pressed at the same time.

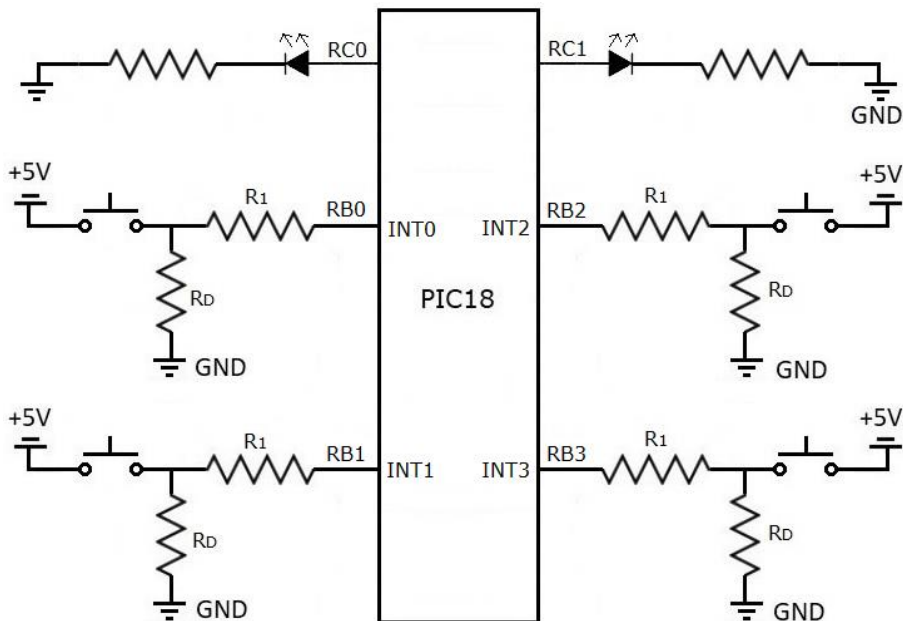


Fig P7.9 Button/Switch & its LED Interfacing

	#INCLUDE <P18F8720.INC>; CAN BE OTHER PROCESSOR		
	ORG	0x00	; Set Origin 0x0000 of Flash memory
	GOTO	START	; Go to Start of Main program.
	ORG	0x08	; Set Origin 0x08 for High Priority Interrupt ; and 0x18 for Low Priority Interrupt

	GOTO	X_ISR	; Go to X_ISR
	ORG	0x100	; Set Origin 0x100 for RB_ISR
X_ISR	BTFSS	INTCON,INT0IF	; Check if interrupt is due to INT0IF, Skip next ; if its due to INT0IF go to X0_ISR
	GOTO	IL1	; if not due to INT0IF go to IL1
	GOTO	X0_ISR	; Go to X0_ISR
IL1	BTFSS	INTCON3,INT1IF	; Check if interrupt is due to INT1IF, Skip next ; if its due to INT1IF go to X1_ISR
	GOTO	IL2	; if not due to INT1IF go to IL2
	GOTO	X1_ISR	; Go to X1_ISR
IL2	BTFSS	INTCON3,INT2IF	; Check if interrupt is due to INT2IF, Skip next ; if its due to INT2IF go to X2_ISR
	GOTO	IL3	; if not due to INT2IF go to IL3
	GOTO	X2_ISR	; Go to X2_ISR
IL3	BTFSS	INTCON3,INT3IF	; Check if interrupt is due to INT3IF, ; Skip next if its due to INT3IF go to X3_ISR
	RETFIE		; If not due to any external interrupts, ; return from interrupt.
	GOTO	X3_ISR	; Go to X3_ISR
	ORG	0x200	; Set Origin 0x200 for Xx_ISRs
X0_ISR	BCF	PORTC,0	; Set for Button at Pin 0, LED 0 OFF
	BCF	PORTC,1	; Set for Button at Pin 0, LED 1 OFF
	BCF	INTCON,INT0IF	; Clear INT0IF flag
	RETFIE		; Return from fast Interrupt Enable
X1_ISR	BSF	PORTC,0	; Set for Button at Pin 1, LED 0 ON
	BCF	PORTC,1	; Set for Button at Pin 1, LED 1 OFF
	BCF	INTCON3,INT1IF	; Clear INT1IF flag
	RETFIE		; Return from fast Interrupt Enable
X2_ISR	BCF	PORTC,0	; Set for Button at Pin 2, LED 0 OFF
	BSF	PORTC,1	; Set for Button at Pin 2, LED 1 ON
	BCF	INTCON3,INT2IF	; Clear INT2IF flag
	RETFIE		; Return from fast Interrupt Enable
X3_ISR	BSF	PORTC,0	; Set for Button at Pin 3, LED 0 ON
	BSF	PORTC,1	; Set for Button at Pin 3, LED 1 ON
	BCF	INTCON3,INT3IF	; Clear INT3IF flag
	RETFIE		; Return from fast Interrupt Enable
	ORG	0x300	; Set Origin 0x300 for Main program
START	SETF	TRISB,0	; Set all pins of PORT B as I/Ps, TRISB = 0xFF
	CLRF	TRISC,0	; Set all pins of PORT C as O/Ps, ; TRISC = 0x00

	BSF	RCON,IPEN	; Enable Priority levels
	MOVLW	0xFA	; WREG = 0xFA, ; RBPU,INTEDG0,INTEDG1,INTEDG2, ; IEDG3,-,INT3IP,-
	MOWF	INCON2	; INTCON2 = WREG = 0xFA
	MOVLW	0xF8	; WREG = 0xF8, ; INT2IP,INT1IP,INT3IE,INT2IE, ; INT1IE,INT3IF,INT2IF,INT1IF
	MOWF	INCON3	; INTCON3 = WREG = 0xF8
	MOVLW	0x90	; WREG = 0x90, GIE,-,INT0IE,-,INT0IF,-
	MOWF	INCON	; INTCON = WREG = 0x90
M1	NOP		; No Operation, For Alternate Logic
	BRA	M1	; Branch to M1 to repeat, Wait on INTO
	END		